Intergenerational Transmission of Victimization*

Sonia Bhalotra† University of Warwick

N. Meltem Daysal‡ University of Copenhagen

Mathias Fjællegaard Jensen§ University of Oxford

Thomas H. Jørgensen¶ University of Copenhagen

Sébastien Montpetit‖ University of Warwick

May 20, 2026

Abstract

Using four decades of Danish administrative data, we estimate the intergenerational transmission of violent crime victimization. Sons are twice as likely, and daughters three times as likely, to be victimized if a parent was victimized, with stronger associations if the mother was the victim. Controlling for cohort, municipality, socioeconomic factors, parental cohabitation, and parental offending explains about 60% of this correlation. The link is weaker in higher-income families; it persists for sons, but is driven to zero for daughters. Further, children of victimized parents experience lower absolute income mobility, comparable to the Black-White difference for men in the United States.

Keywords: victimization, violent crime, intergenerational transmission, income mobility
JEL Classifications: K42, J12, J62

*We would like to thank Randi Hjalmarsson for helpful discussions. Bhalotra and Montpetit acknowledge funding from ERC Advanced Grant number 885698 (Evidence-VAW). Jørgensen acknowledges financial support from the Carlsberg Foundation, grant CF22-0317. The Center for Economic Behavior and Inequality (CEBI) at the University of Copenhagen is financed by grant DNRF134 from the Danish National Research Foundation.

†Department of Economics, University of Warwick, CAGE, IFS, CESifo, RFBerlin, IZA, AcSS, IEA (email: sonia.bhalotra@warwick.ac.uk)

‡Department of Economics, University of Copenhagen, CEBI, CESifo, IZA, RFBerlin (email: meltem.daysal@econ.ku.dk)

§Department of Economics, University of Oxford (email: mathias.jensen@economics.ox.ac.uk)

¶Department of Economics, University of Copenhagen, CEBI (email: thomas.h.jorgensen@econ.ku.dk)

‖Department of Economics, University of Warwick (email: sebastien.montpetit@warwick.ac.uk)

1 Introduction

A substantial fraction of the population experiences violent crime victimization. In the United States, an estimated 2.3% of the population were victims of violent crime in 2022 alone.[1] Beyond its immediate physical and psychological harms, a growing body of evidence demonstrates that violent crime victimization also entails severe and persistent economic consequences (Chang et al., 2025; Adams et al., 2024a,b; Dustmann and Mertz, 2024; Bindler and Ketel, 2022). Understanding the determinants of victimization is therefore important not only from a public health perspective, but also from an economic one. While a large literature documents the intergenerational transmission of criminal behavior, dating back to the nineteenth century (Dugdale, 1877) and surveyed in Besemer et al. (2017) and Wildeman (2020), we know substantially less about the family origins of victimization and its intergenerational transmission.[2]

In this paper, we estimate intergenerational correlations in violent crime victimization between parents and children, separately for fathers and mothers and for sons and daughters. Theories across disciplines highlight complementary mechanisms behind the intergenerational transmission of victimization and violence. In economics, the framework from Becker and Tomes (1979) highlights how parental resource constraints—which existing evidence suggests are tightened following victimization—can lower investments in child safety and stability (Van der Kolk, 1989; Bowlby, 1979; Bandura and Walters, 1977). Sociological theories of social disorganization emphasize how neighborhood conditions—concentrated poverty, weak institutions, and residential instability—shape exposure to violence (Sampson and Laub, 1997; Shaw and McKay, 1942).[3] Psychological theories stress that early exposure to violence may generate persistent vulnerability through social learning, impaired socio-emotional functioning, and maladaptive coping. Motivated by these theories, we explore several potential mechanisms, including parental socioeconomic status and location effects.

A central challenge in estimating the intergenerational correlations in victimization is the identification of victimization across multiple generations. While governments have long sought to systematically record criminal behavior, information on victims is often incomplete or unavailable. In Denmark, for instance, police registers on violent victimization have been available only since 2001, providing insufficient temporal depth to study parent–child correlations. We address this limita-

[1] https://bjs.ojp.gov/document/cv22.pdf, last retrieved April 27, 2026.

[2] One possible reason is that criminal behavior is more readily conceptualized as an economic decision, whereas victimization is often perceived as something that merely "happens to people," potentially at random. While victims are never to blame, individuals nevertheless exhibit some agency in shaping their exposure to victimization risk, which has been shown to vary systematically with age (Bindler et al., 2024), alcohol consumption (Chalfin et al., 2023), and the surrounding social environment (Lindo et al., 2018; Anderson et al., 2013; Card and Dahl, 2011; Rees and Schnepel, 2009).

[3] Related work in economics highlights neighborhood effects on youth crime and intergenerational mobility (Finlay et al., 2023; Chetty and Hendren, 2018a,b; Chetty et al., 2016; Kling et al., 2005; Ludwig et al., 2001).

tion by leveraging Danish administrative medical records, which document the causes of injuries at the time of hospital contact, including assault-related injuries. These records span more than four decades, allowing us to identify violent crime victimization for both parents and children. Our main analysis sample consists of more than 370,000 children born between 1980 and 1987 and their family members, whom we link across medical, crime, and labor market registers.

Our main findings are as follows. First, we find large unconditional correlations of parent-child victimization. Sons of victimized parents are almost twice as likely to experience an assault-related ER contact as sons of parents who were not victimized. The difference is larger for daughters, their risk being more than three times the baseline risk. The mother's victimization has a larger association with the victimization risks of both sons and daughters, and more so for daughters. The magnitudes involved in the intergenerational transmission of victimization are broadly similar to those documented in the more comprehensively studied intergenerational transmission of perpetration, which are, in turn, similar to the magnitudes for the intergenerational transmission of income (Hjalmarsson and Lindquist, 2012). Thus, our results establish that family persistence in victimization risk is a first-order phenomenon, worthy of further analysis.

Our second main result is that, while adjusting for cohort and municipality-of-birth fixed effects does not make much difference to the correlation, controlling for measures of parent socioeconomic status (SES), for parent cohabitation, and for the parent being a perpetrator explains about 50% to 70% of the victimization correlation. These observables explain a larger share of the association driven by fathers than is the case for mothers. The remaining 30% to 50% of the variation is attributable to unobservables, which plausibly include social learning and the related facets of family-level exposure that the psychology literature has emphasized.

Our third key finding is that the correlation of parent-child victimization is attenuated for families in the upper part of the SES distribution. This pattern is clearer for daughters than for sons. We find no detectable difference by whether the child was living with the victimized parent when the parent was victimized, or by the age at which the child is exposed to parental victimization.[4]

Fourth, we document cross-type intergenerational links between perpetration and victimization. While the intergenerational association is generally stronger within type, a notable exception is sons exposed to a violent father, for whom the effect on victimization risk exceeds that of a victimized father and approaches the effect of a victimized mother. These cross-type patterns suggest that perpetration and victimization share overlapping but not identical transmission channels.

Our fifth contribution is to provide the first estimates of the relationship between violence victimization and intergenerational income mobility. We find that children exposed to violence experience significantly lower levels of absolute income mobility, around 8.5 and 5.8 rank points for sons and daughters, respectively. These mobility gaps cannot be fully explained by other parental

[4]However, we do not observe parental victimization for children younger than age 8; see Section 3.

characteristics such as parents' socioeconomic status and marital stability.

In order to benchmark the victimization results, we also estimate intergenerational transmission in two closely related domains: violent perpetration and domestic violence victimization. The estimates fall broadly within the range of available evidence. We find that sons of violent fathers are about 20 percentage points more likely to be convicted of a violent offense, an estimate that falls to about 9 percentage points after adding controls. For domestic violence, daughters of mothers who experienced domestic violence (DV) are 8.7 percentage points more likely to experience DV themselves, with the estimate falling to 3.3 percentage points conditional on controls.

The intergenerational correlations in victimization likely reflect a mix of causal pathways—such as parental earnings losses following victimization (Bindler and Ketel, 2022), role-model effects, and impacts on parenting styles—alongside exposure to risk factors common to parents and children, including poverty, ethnicity, and neighborhood. Distinguishing between these channels in future work is essential for designing policies aimed at disrupting the observed cycle of victimization across generations.

Contributions to existing research. To our knowledge, this is the first large-scale study to estimate intergenerational transmission of victimization, violent perpetration, and domestic violence, as well as to quantify a common set of potential drivers in a single setting using population-representative administrative data.

Our primary contribution is to research on the causes of victimization. Existing research has identified a role for risky behaviors, leveraging age-thresholds in policies including the minimum legal drinking age and the minimum age for school dropout (Bindler et al., 2024; Chalfin et al., 2023; Anderson et al., 2013), and a role for socioeconomic status, suggesting that family income is protective (Dustmann and Mertz, 2024). We provide the first estimates of the influence of parental assault victimization, considering all assaults.

Related research with focus on DV has noted a mother-daughter correlation in victimization, anticipated in Pollak (2004).[5] Using data on assaults identified as having occurred in a residential setting, we approximate DV. Our results for DV contribute to the DV literature in several ways. First, we show that the mother-daughter correlation is eliminated at higher levels of parental income. Second, we use full population administrative data that include fathers, and that model not only the parent's victimization but also the parent's perpetration. This is important, given evidence that child witnesses of adult DV were more likely to be both victims and perpetrators in their adolescent relationships (Forke et al., 2018). Third, we use an objective measure of DV while most other research uses self-reported DV. This is relevant given evidence that DV is under-reported and that there is

[5] See, for instance, Puno et al. (2023), Madruga et al. (2017), and Wood and Sommers (2011), and recent work in economics which uses self-reported indicators of own and parental intimate-partner violence (Collins, 2025; Hernandez-Leal et al., 2025).

selection into reporting, which plausibly varies across generations.

We additionally contribute to research on intergenerational mobility. A number of studies document intergenerational correlations in income and education (Chetty et al., 2014; Black and Devereux, 2011; Aaronson and Mazumder, 2008; Behrman and Taubman, 1976). Our estimates for the intergenerational persistence in victimization (and our estimates for perpetration) are broadly similar in size to existing estimates for persistence in income and perpetration.

In addition, we show that parental income is an important mediator of the cycle of violence, and that exposure to parental victimization is associated with reduced intergenerational income mobility.

Outline. The rest of the paper is structured as follows. Section 2 and Section 3 describe the empirical strategy and the data used, respectively. Section 4 provides our empirical results on the cycle of violence. Section 5 presents estimates of the association between victimization and social mobility. Finally, Section 6 concludes.

2 Empirical Strategy

Following existing research on intergenerational associations in socioeconomic outcomes, we begin by estimating a simple linear model of violence transmission:

$$Child_i = \gamma_1 + \gamma_2 Parents_i + \eta_i \tag{1}$$

where $Child_i$ and $Parents_i$ denote measures of victimization for children and parents, respectively. The coefficient of interest, γ_2, captures the strength of the intergenerational correlation, providing a benchmark measure of how strongly victimization in one generation predicts victimization in the next. We estimate this model both for the pooled sample and separately by parent–child gender pairs (father–son, father–daughter, mother–son, and mother–daughter) to uncover potential asymmetries in how victimization is transmitted within and across genders.

While this baseline specification documents the raw intergenerational correlation, it is silent on the underlying mechanisms that may account for the observed associations. We therefore turn to specifications that sequentially add a rich set of controls intended to proxy for different pathways of transmission. The purpose of these extended specifications is not to draw causal conclusions about the individual covariates, but rather to assess how much of the observed parent–child association can be explained by observable background factors.

We begin by including child birth-cohort indicators to absorb common temporal shocks, such as changes in policing practices, reporting standards, or national policy reforms. Next, we add municipality-of-birth fixed effects to account for local conditions, including neighborhood disadvantage, social disorganization, and community norms, that may be correlated with both parental

and child outcomes. We then incorporate measures of family stability, using indicators for parental cohabitation and years of cohabitation during the child's upbringing, since instability in family structure may shape both exposure to violence and later risks of victimization or offending. We further control for parental socioeconomic status, measured by income rank within birth cohorts, unemployment duration, and years of schooling. Finally, we include parental criminal activity (violent perpetration, property crime, and driving under the influence of alcohol or drugs) to proxy for parenting behaviors and styles.

To evaluate the relative contribution of these observed factors, we employ Oaxaca-Blinder decompositions. Following Fortin et al. (2011), we define the portion of the gap in child violence victimization between exposed and non-exposed children that is attributable to differences in observed covariates as the "explained gap". The residual "unexplained gap" reflects differences in intercepts and in the returns to parental characteristics across groups.

3 Data and Analysis Sample

3.1 Data

We use population-wide Danish register data covering four decades (1980–2021). These data include individual-level records with unique personal identifiers, which allow us to track individuals over time, link children to parents (regardless of cohabitation), and merge criminal and health records.

Measure of victimization. Victimization is measured using the *National Patient Register*, which records all hospital contacts (inpatient, outpatient, and emergency) across public and private hospitals. For emergency room visits, the primary cause of contact must be specified, with assaults classified as a distinct category. Our main outcome is an indicator for an assault-related ER contact. Parental victimization is measured at child ages 8–17, while child victimization is measured at ages 18–35.[6]

Administrative crime data on victimization are only available from 2001. As noted by Doyle and Aizer (2018), health records may provide a more reliable measure, as police reports may be subject to reporting biases. In Appendix Figure A1, we show that our hospital-based measure captures fewer violent incidents, but follows similar trends over the 2001-2018 period for our cohorts of interest.

Covariates. Demographic characteristics, including child gender, birth year, municipality of residence at birth, and parental marital status, are obtained from the *Population Register*. We draw

[6]Because the emergency room data begin in 1995, we do not observe the relevant patient-register information for children younger than age 8 in the parent cohorts in our sample.

on the *Income Statistics Register* for parental employment and labor income, and on the *Education Register* for years of schooling. Using these data, we construct fixed effects for child birth cohort and municipality of residence at birth. For parents, we construct the following characteristics separately for mothers and fathers: (i) indicators for years of cohabitation during the first 17 years of the child's life, (ii) percentile rank in the national labor income distribution of their child's birth cohort, based on average income during child ages 8–17 (in 2015 DKK) (Chetty et al., 2020), (iii) indicators for years of unemployment during child ages 0–17, and (iv) indicators for years of schooling measured when the child is 17. Parental perpetration is measured using the *Danish Central Crime Register*, which contains complete police and court records for all offenses in Denmark. We focus on cases where formal charges were filed and upheld following investigation.[7] We construct separate indicators of parental perpetration between child ages 8-17 for violent offenses, property crimes, and driving under the influence (DUI) of alcohol or drugs.

Additional outcomes. Building on evidence of intergenerational transmission in crime and DV (Collins, 2025; Hernandez-Leal et al., 2025; Hjalmarsson and Lindquist, 2012), we benchmark the transmission of assault victimization against two outcomes. We use the *Danish Central Crime Register* to study violent perpetration across generations. Because administrative victimization data begin in 2001, we rely again on Danish hospital records to measure DV. For assault-related ER contacts, physicians are recommended to record whether the incident occurred in a residential area, so we define DV as assault-related victimization recorded in a residential setting.[8]

3.2 Analysis Sample and Descriptive Statistics

Our analysis focuses on children born between 1980 and 1987, allowing us to track outcomes up to age 35. We exclude children born abroad and those who cannot be linked to parents, and we restrict our attention to observations with non-missing covariates. The final sample includes 370,041 children with linked parental records.

Table 1 reports descriptive statistics.[9] Panel A focuses on our key outcome, assault victimization, while Panel B turns to our proxy for DV. Columns (1)–(3) present outcomes for children observed between ages 18 and 35, both for the full sample and separately by exposure to parental violence. Columns (4)–(5) report corresponding measures for parents, recorded at child ages 8–17.

The statistics in Panel A indicate that approximately 3% of children were exposed to parental victimization during ages 8–17. Fathers are more likely to be victimized than mothers (2.03% vs

[7] In Denmark, arrests are relatively rare compared to the United States; charges filed and upheld provide the closest equivalent measure.

[8] We recognize that this is a proxy that could include a wider set of cases. Assault could, for instance, result from a fight between two men, or two women. Among female victims, domestic violence is the more likely cause.

[9] Appendix Table C1 provides summary statistics for violent perpetration.

1.29%). There is a large intergenerational gradient: children exposed to parental victimization have a 19.16% assault-victimization rate compared with 8.93% among the non-exposed, roughly a doubling of risk. There is a clear gender asymmetry. Sons have higher victimization rates in levels (13.81% vs 4.56% for daughters), but the proportional increase associated with parental exposure is steeper for daughters, rising from 4.27% to 13.21% (about threefold) compared with sons' increase from 13.45% to 24.98% (about twofold).

Table 1: Descriptive Statistics

		Child victimization by parental victimization		Parental victimization		
	Full sample	Exposed	Non-exp.	Father	Mother	Exposed children
Assault Victimization						
All children	9.26	19.16	8.93	2.03	1.29	11,758
	(28.99)	(39.36)	(28.52)	(14.10)	(11.28)	
Sons	13.81	24.98	13.45	1.98	1.31	5,756
	(34.50)	(43.29)	(34.12)	(13.94)	(11.37)	
Daughters	4.56	13.21	4.27	2.08	1.27	6,004
	(20.86)	(33.86)	(20.22)	(14.26)	(11.19)	
Domestic Violence Victimization						
All children	2.90	9.91	2.79	0.71	0.90	5,784
	(16.79)	(29.88)	(16.47)	(8.41)	(9.44)	
Sons	3.60	10.38	3.49	0.70	0.91	2,855
	(18.64)	(30.50)	(18.36)	(8.37)	(9.52)	
Daughters	2.18	9.42	2.06	0.72	0.89	2,928
	(14.60)	(29.21)	(14.21)	(8.47)	(9.37)	

Notes: The data source is the Danish population (BEF) and the hospital admission register (LPR-ADM). We proxy for domestic violence victimization with assaults occurring in a residential area. Victimization outcomes are observed at child age 8-17 for parents, and child age 18-35 for children. Statistics are rescaled by 100 to reflect percentages. The last column reports the number of children exposed to at least one victimized parent in a given exposure-child gender cell. The number of observations is 370,041 for the full sample; 188,010 for sons; and 182,031 for daughters.

Turning to Panel B, because our DV proxy is a subset of the broader assault measure by construction, exposure to DV is mechanically less common than exposure to any parental victimization, at 1.56% compared with 3.18% in Panel A. Among children aged 18–35, prevalence similarly declines from general assault victimization at 9.26% to DV contacts at 2.90%. Intergenerational transmission remains strong. Among children exposed to DV, their own risk of DV rises from 2.79% to 9.91% (about a 3.5-fold increase). As with overall assault victimization, levels of victimization are higher

among sons than daughters, but proportional transmission is consistently steeper for daughters. Overall, the descriptive evidence indicates a sizeable intergenerational correlation in victimization, with meaningful gender differences.

4 The Cycle of Victimization

Figure 1 displays the intergenerational correlations in assault victimization. While the descriptive statistics in Section 3 preview these correlations, we now quantify them using equation (1) and assess their robustness to the sequential inclusion of controls. The top panel shows transmission from parents to sons, and the bottom panel shows transmission to daughters. Within each panel, we distinguish between exposure from any parent, exposure from the father, and exposure from the mother. The first row in each panel reports unconditional correlations, while rows (2)–(6) present results from specifications that sequentially add additional controls.[10]

Unconditional correlations. The unconditional correlations in row (1) of Figure 1 show strong intergenerational transmission of victimization. Sons of exposed parents are 11.53 percentage points (pp) more likely to experience an assault-related ER contact, nearly doubling their baseline risk of 13.45%. For daughters (Panel b), the proportional increase is even larger: parental victimization raises their likelihood of an assault-related ER contact by 8.94 pp, a more than threefold increase relative to the 4.27% baseline.

Disaggregating by parent gender reveals that transmission from mothers is stronger for both sons and daughters. For sons, paternal victimization raises the risk by 10.69 pp, while maternal victimization increases it by 13.56 pp. For daughters, the maternal channel is even stronger. Daughters of victimized mothers are 11.53 pp more likely to be victimized, a 260% increase over the baseline, compared to 7.7 pp for paternal victimization.

Role of mediating factors. Rows (2)–(6) assess how the transmission coefficients change when we sequentially add controls. We find that adding cohort and municipality fixed effects has little impact on the strength of the transmission for either sons or daughters, indicating that broad time and place factors are not first-order drivers. The estimates attenuate substantially once family stability and parental SES are taken into account. Controlling for parental criminal behaviors, measured by indicators for involvement in violent, property, or DUI offenses, further reduces the magnitude of the correlation, consistent with behavioral factors correlated with parenting styles and household environments being important channels behind the intergenerational persistence of victimization.

[10] Appendix Table A1 and Appendix Table A2 report the corresponding regression estimates.

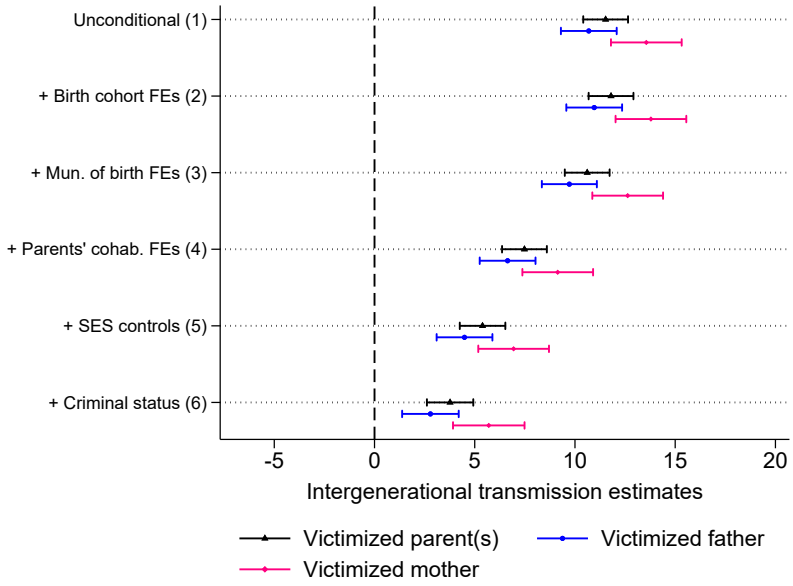
Moving from the unconditional to the fully saturated specification reduces the transmission from any parent from 11.53 to 3.77 pp for sons and from 8.94 to 3.59 pp for daughters. The attenuation is more pronounced for paternal transmission: father-to-son estimates fall from 10.69 to 2.79 (a 74% reduction) and father-to-daughter from 7.70 to 2.27 (a 71% reduction), whereas maternal links remain larger after adjustment, declining from 13.56 to 5.70 for sons (a 58% reduction) and from 11.53 to 5.80 for daughters (a 50% reduction).

To gauge how much of the intergenerational link is accounted for by these observables, we implement Oaxaca–Blinder decompositions (see Appendix Table A3 and Appendix Table A4). For transmission from any parent, the included covariates jointly explain about 62% of the raw gap for sons and 56% for daughters, leaving 38% and 44% unexplained, respectively. For transmission from fathers, the explained share is somewhat higher, around 67% for sons and 66% for daughters, while for mothers, the explained share is smaller—approximately 56% and 48%—thus maternal pathways remain more persistent after accounting for the same set of covariates.

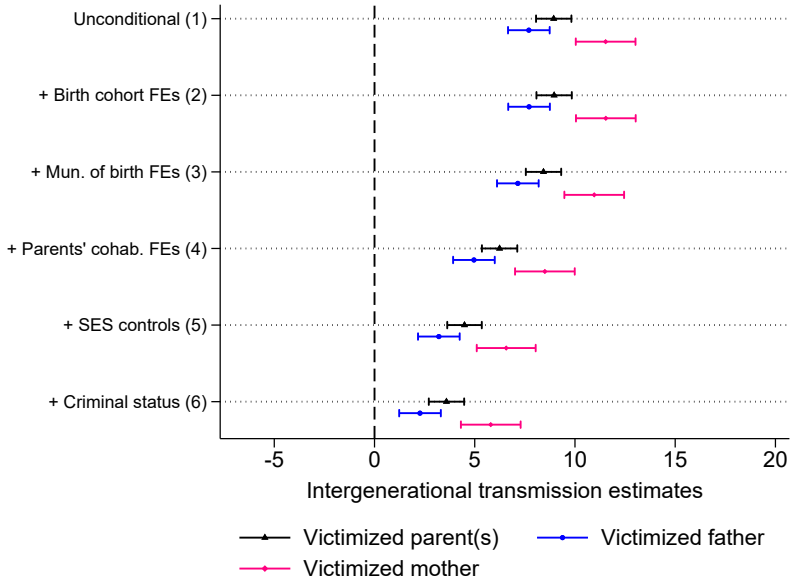
Taken together, the results reveal that while family structure, socioeconomic resources, and parental behaviors absorb a substantial portion of the intergenerational association, the transmission coefficients remain statistically and economically significant. A large residual correlation persists, especially along maternal lines, suggesting that unobserved family dynamics, parenting styles, or direct causal mechanisms contribute to shaping the intergenerational persistence of victimization beyond the measured mediators.

Heterogeneous effects. A large body of research demonstrates that shocks experienced during childhood have persistent effects on later-life outcomes, with growing evidence that developmental trajectories remain malleable well beyond early childhood (Bhalotra et al., 2026; Cunha and Heckman, 2007; Carneiro and Heckman, 2003). Building on this literature, we examine two potential dimensions of heterogeneity in transmission using the fully saturated specification in row (6) above. First, we distinguish between parental victimization experienced during middle childhood (ages 8–12) and adolescence (ages 13–17),[11] replacing the single indicator for parental victimization with two period-specific indicators in the fully saturated specification. In line with Carneiro et al. (2021), the coefficients are larger for opposite-gender parent–child pairs during the teenage years, but the differences are not statistically significant (columns 1, 4, 7, and 10 of Table 2). Second, we examine whether transmission operates through children witnessing the assault by interacting parental victimization with an indicator for co-residence at the time of the assault-related ER contact. The interaction terms (columns 2, 5, 8, and 11 of Table 2) are positive. However, they are not statistically significant, which we attribute to limited power given that few children co-resided with their fathers at the time of the incident.

[11] Fewer than 9% of exposed children experience parental victimization in more than one developmental period.



(a) Assault victimization, sons



(b) Assault victimization, daughters

Figure 1: Intergenerational transmission of assault victimization

Note: These figures report estimates of violence transmission (γ_2) using equation (1). Victimization outcomes are observed at child age 8-17 for parents, and child age 18-35 for children. The dependent variable is whether the child appears in the emergency-room hospitalization data for assault between the ages of 18 and 35. Income ranks (0-100) are determined within child birth-year cohorts, regardless of victimization status. Point estimates (and standard errors) are rescaled by 100 to ease interpretation and readability. Robust standard errors in parentheses. 95% confidence intervals are reported.

Table 2: Heterogeneous transmission of assault victimization

Dependent variable: Child victim$_i$												
Transmission from:	Mother						Father					
Child gender:	Daughters			Sons			Daughters			Sons		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Middle childhood$_i$	7.33***			3.61*			1.53			3.16**		
(8-12 years old)	(1.43)			(1.51)			(0.87)			(1.22)		
Adolescence$_i$	5.74***			6.58***			2.75***			3.70***		
(13-17 years old)	(0.90)			(1.15)			(0.71)			(0.94)		
Victimized parent$_i$		4.93***	6.89***		5.19***	5.85***		2.32***	2.52***		2.57***	2.64**
(8-17 years old)		(1.02)	(0.89)		(1.26)	(1.05)		(0.54)	(0.63)		(0.74)	(0.83)
Victimized parent$_i$			-5.68***			-0.73			-1.12			0.65
× Above med. income$_i$			(1.50)			(2.09)			(1.08)			(1.62)
Victimized parent$_i$		1.67			1.02			-1.15			4.07	
× Lived with parent(s)		(1.49)			(1.78)			(2.59)			(3.22)	
Parental controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean non-exposed	4.41	4.41	4.41	13.63	13.63	13.63	4.40	4.40	4.40	13.60	13.60	13.60
Observations	182,031	182,031	182,031	188,010	188,010	188,010	182,031	182,031	182,031	188,010	188,010	188,010
R-squared	0.030	0.030	0.030	0.028	0.028	0.028	0.030	0.030	0.030	0.028	0.028	0.028
p-value of $\gamma_2 + \gamma_3 = 0$		0.00	0.32		0.00	0.00		0.65	0.11		0.03	0.02
p-value of $\gamma_2 - \gamma_3 = 0$	0.34			0.11			0.27			0.72		
N, 8-12-year-olds	696			811			1,313			1,229		
N, 13-17-year-olds	1,552			1,526			2,113			2,137		
N, 8-17-year-olds		2,308	2,308		2,464	2,464		3,778	3,778		3,730	3,730
N, 8-17-year-olds, above median			424			485			777			761
N, 8-17-year-olds, cohabiting		1,203			1,250			159			202	

Notes: Victimization outcomes are observed at child ages 8-17 for parents, and child ages 18-35 for children. The dependent variable is whether the child appears in the emergency-room hospitalization data for assault victimization between the ages of 18 and 35. Income ranks (0-100) are determined within child birth-year cohorts, regardless of victimization status. In columns with the above-median-income or co-residence interactions, the row labelled p-value of $\gamma_2 + \gamma_3 = 0$ tests whether transmission coefficients are statistically different from zero for above-median-income or co-residing children. In columns with age-specific exposure indicators, the row labelled p-value of $\gamma_2 - \gamma_3 = 0$ tests equality of the middle-childhood and adolescent exposure coefficients. Point estimates (and standard errors) are rescaled by 100 to ease interpretation and readability. Robust standard errors in parentheses. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Given evidence that parental resources shape child development and buffer the effects of adverse shocks (Aizer and Cunha, 2023; Dahl and Lochner, 2012), we examine whether intergenerational transmission varies by family SES. Columns (3), (6), (9), and (12) of Table 2 present estimates from models that include an interaction term between parental victimization and an indicator for above-median household income. The results show that transmission is weaker among higher-income families, particularly for daughters. The interaction coefficients are negative in all specifications and sizeable in magnitude, roughly equal to the main effect for maternal victimization and about half as large for paternal victimization, and precisely estimated for the maternal case. For mothers, we cannot reject the hypothesis that income fully mediates the intergenerational link in victimization ($p = 0.32$), while for fathers the corresponding test ($p = 0.11$) suggests borderline significance, and we see some attenuation, without income fully eliminating the correlation. By contrast, transmission to sons does not vary systematically with parental income rank, implying that socioeconomic resources play a more important mitigating role for daughters than for sons.

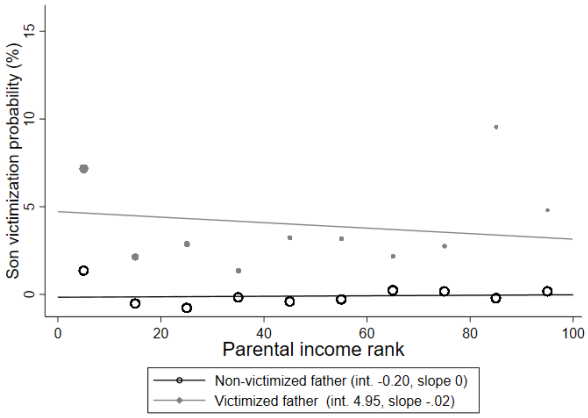
Figure 2 provides complementary non-parametric evidence on this income gradient. The figure plots bin-scatters of residualized parental income against child victimization, using the fully saturated specification. The intergenerational gap in victimization closes almost entirely at the top of the income distribution for daughters and is markedly reduced for mother–son pairs, while it remains more persistent for father–son pairs. Overall, these results suggest that income support and broader resource-based policies may be effective in weakening this transmission across generations, particularly for daughters.

4.1 Benchmarking Victimization Against Other Forms of Violence

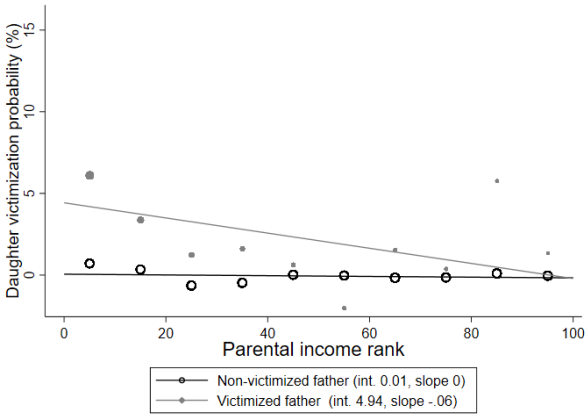
In Appendix B and Appendix C, we replicate our main analysis of intergenerational victimization using two related outcomes, domestic violence and violent perpetration. We highlight selected results here because they provide useful benchmarks for our main findings.

Appendix Figure A2 summarizes how our estimates, expressed as odds ratios, compare with existing estimates (a detailed comparison is in Appendix Table C7). Overall, for both DV victimization and violent perpetration, the conditional results fall in the range of estimates observed in the literature. However, our unconditional estimates based on our measure of assaults in hospital records exceed estimates in the literature. It is notable that, whether conditional or unconditional, our estimates for Denmark are not lower than estimates from developing countries.

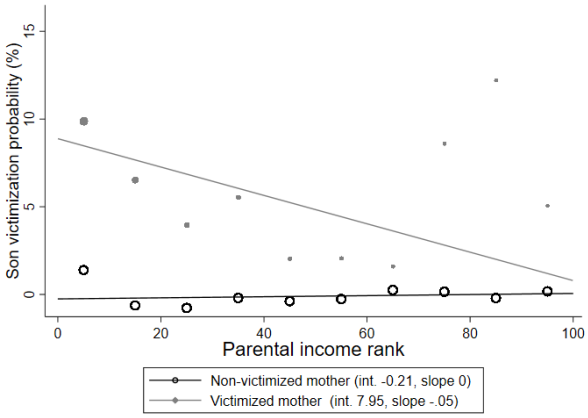
Motivated by the literature on the intergenerational transmission of crime, which largely focuses on fathers and sons (e.g. Wildeman, 2020; Besemer et al., 2017; Hjalmarsson et al., 2015), Appendix Table C2, Panel B shows that sons of violent fathers are about 20 percentage points more likely to be convicted of a violent offense, almost trebling the baseline risk of 8.2%. Adding controls reduces



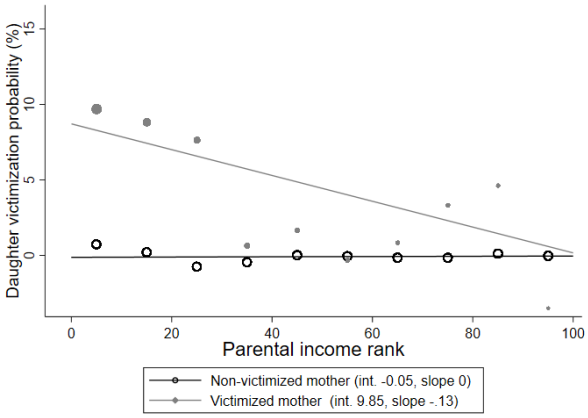
(a) Fathers to Sons



(b) Fathers to Daughters



(c) Mothers to Sons



(d) Mothers to Daughters

Figure 2: Child victimization outcomes across parental income deciles by parental victimization.

Note: Each data point reports the child victimization rate at a given decile of the parental income distribution, residualized with respect to all controls other than parental income. Victimization outcomes are observed at child age 8-17 for parents, and child age 18-35 for the children. The dependent variable is whether the child appears in the emergency-room hospitalization data due to an assault between the ages of 18 to 35. Income ranks, 0-100, are determined within child birth-year cohorts, regardless of victimization status. Cells with low counts are not reported for confidentiality reasons. Symbol sizes are weighted by the number of observations.

the estimated treatment effect of parental offending by about half, to roughly 9 percentage points. This is similar to, for example, the conditional correlation in Hjalmarsson and Lindquist (2012) who find that sons of convicted fathers have 2.1 times higher odds of conviction by age 31 than sons of non-offenders.

Appendix Table B2, Panel C turns to DV between mothers and daughters, motivated by the IPV literature, which documents elevated risks among daughters whose mothers experienced IPV (Collins, 2025; Hernandez-Leal et al., 2025; Puno et al., 2023; Madruga et al., 2017; Islam et al., 2014; Wood and Sommers, 2011). Daughters whose mothers experienced domestic violence are 8.7 percentage points more likely to experience domestic violence themselves. With controls, the estimate falls to 3.3 percentage points, still a 2.5-fold increase relative to the 2.1% baseline risk. The latter magnitude closely matches the existing DV literature, although that literature typically relies on survey reports and has been more focused on developing countries.[12]

These estimates help benchmark our main victimization results. Among fathers and sons, the unconditional victimization estimate of 10.7 percentage points is about half as large as the corresponding estimate for violent perpetration, 19.8 percentage points. Both estimates attenuate with controls, but more strongly for victimization than for perpetration. The adjusted coefficients, 2.8 and 8.9 percentage points, suggest that the father-son association in victimization is more closely tied to shared socioeconomic and environmental factors, whereas the persistence of violent offending extends further beyond observables.

Among mothers and daughters, the pattern is more similar across outcomes. The unconditional victimization estimate, 11.5 percentage points, slightly exceeds the corresponding domestic violence estimate of 8.7 percentage points, and both fall by about half once controls are introduced. The adjusted coefficients, 5.8 and 3.3 percentage points, imply sizable and similarly persistent intergenerational associations.

From perpetration to victimization. In Appendix Table C5 and Appendix Table C6, we also document that a cycle of violence exists between types of exposure to violence. The intergenerational association is stronger within type—from parental victimization to child victimization and from parental perpetration to child perpetration—with the notable exception of sons exposed to a violent father. Exposure to a violent father increases sons' victimization risk more than exposure to a victimized father, and almost as much as exposure to a victimized mother.

Overall, these comparisons suggest that parental victimization reproduces vulnerability across generations through mechanisms that overlap with, but are not limited to, those underlying the

[12] Appendix Table B1 and Appendix Table B2 report the corresponding estimates for assault victimization in residential areas for sons and daughters, respectively. Appendix Table C2 and Appendix Table C3 report the analogous estimates for violent perpetration.

transmission of violent behavior and domestic violence.

5 Victimization and Social Mobility

Interactions of income and victimization can also manifest in intergenerational income transmission. In this section, we evaluate how intergenerational income mobility differs between children of victimized parents and those who grew up in non-victimized households. We follow the popular approach to studying absolute and relative income mobility, namely the rank-rank relationship.[13] The standard model regresses the child's rank in the income distribution on that of the parents. We adopt the same strategy as in previous work on heterogeneity in absolute and relative mobility by allowing for differences in intercepts and interacting parents' income rank with the characteristic of interest (e.g. Boustan et al., 2025; Asher et al., 2024; Abramitzky et al., 2021; Alesina et al., 2021; Chetty et al., 2020). Specifically, we estimate the following model:

$$Child\ rank_i = \alpha_1 + \alpha_2\, Victim_i + \beta_1\ Parents\ rank_i + \beta_2\ Parents\ rank_i \times Victim_i + \varepsilon_i \quad (2)$$

where $Child\ \ rank_i$ is child i's rank in the child income distribution. Parental victimization, $Victim_i$, is a dichotomous variable taking the value of one if at least one of i's parents is a victim of assault. We refer to α_1 as the level of absolute mobility (i.e. the predicted income rank for those with parents at the bottom of the income distribution) for children not exposed. $1-\beta_1$ is the level of relative mobility, i.e. the extent to which children's income rank is related to parental income ranks. The main coefficients of interest are α_2 and β_2. α_2 indicates the difference in a child's predicted income rank among children of parents with the lowest income if the child has a victimized parent (i.e. the difference in absolute mobility). β_2 measures the differences in relative mobility between children exposed to parental victimization and those who are not.

Baseline specification. We first estimate equation (2) without additional covariates using our two-generation sample, separately for sons and daughters. Results are reported in columns 1 and 3 of Table 3.

We find that children of victimized parents display significantly lower rates of absolute income mobility (α_2 is negative). The predicted child's income rank among the lowest income households with a victimized parent is 8.49 and 5.82 points lower for sons and daughters, respectively. These estimates are sizable, as they suggest 19% lower absolute income mobility among exposed children (boys and girls alike), similar to the Black-White difference for men in the United States (Chetty et al., 2020). Because parental victimization is more prevalent in the lower part of the parental income

[13] See Landersø and Heckman (2017) and Eshaghnia et al. (2022) for analyses of social mobility in Denmark comparing across measures of mobility.

Table 3: Social mobility by parental violence victimization and child gender

Dep. var: child's income rank				
	Daughters		Sons	
	(1)	(2)	(3)	(4)
Victimized parent$_i$ (α_2)	-5.816***	-1.894***	-8.493***	-1.656***
	(0.474)	(0.484)	(0.606)	(0.593)
Parents' rank$_i$×	0.066***	0.028**	0.069***	-0.008
Victimized parent$_i$ (β_2)	(0.013)	(0.013)	(0.015)	(0.015)
Parents' rank$_i$ (β_1)	0.267***	0.151***	0.255***	0.155***
	(0.003)	(0.002)	(0.003)	(0.004)
Constant (α_1)	30.14***	25.39***	44.21***	33.58***
	(0.128)	(5.595)	(0.151)	(10.110)
Parental controls	No	Yes	No	Yes
Observations	182,031	182,031	188,010	188,010
R-squared	0.086	0.110	0.063	0.087

Notes: This Table reports results of rank-rank regressions allowing for heterogeneous effects of parental income by parental violence victimization following equation (2). Victimization outcomes are observed at child age 8-17 for parents. The dependent variable is the child's income rank where its income is averaged over ages 30-35. Income ranks, 0-100, are determined within child birth-year cohorts, regardless of victimization status. Point estimates (and standard errors) are rescaled by 100 to ease interpretation and readability. Robust standard errors in parentheses. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

distribution, these differences are economically relevant and may also predict the child's own victimization risk, other things equal. However, parental income acts as a strong mediator of the effect of victimization on the child's income. In other words, relative mobility ($-\beta_2$) is lower (the rank-rank relationship steeper) for exposed children.

Comparing across child gender reveals important differences. For daughters of parents at the top of the income distribution, the linear model predicts no difference in mobility by parental victimization ($\alpha_2 + 100\beta_2 \approx 0$). In contrast, the social mobility gap between exposed and non-exposed sons does not fully close at the top, where a difference of more than one rank point persists. To further illustrate this result, we report these regression estimates in Appendix Figure A3 along with rank-rank associations along the parental income distribution. Consistent with the results of the linear model, we observe a persistent gap in intergenerational income mobility for sons, but not for daughters. We remain, nevertheless, cautious in interpreting the non-linear relationships, given that we have little statistical power to assess the extent of mobility gaps at the top of the income distribution, where victimization is less prevalent.

Role of mediating factors. In even columns of Table 3, we control for the full set of parental characteristics as in row 6 of Figure 1. We focus on exposure to victimization of any parent as we have little statistical power when focusing on each parent separately.[14]

We find that parental controls explain most of the gaps in absolute income mobility between exposed and non-exposed children (α_2 shrinks), similar to Jensen and Manning (2025) who find that the higher mobility rates of children of immigrants relative to children of locals in Denmark are largely explained by differences in other background characteristics. In our case, significant residual gaps remain for both sons and daughters. For sons it is notable that the gap persists at the top of the parental income distribution.

To assess the relative role of parental income in explaining mobility gaps by exposure to victimization, we again employ Oaxaca-Blinder decompositions. We decompose the child income gap between exposed and non-exposed children into a portion explained by parental income alone, a portion explained by the full set of parental characteristics, and an unexplained residual. The results, reported in Appendix Table A7, reveal that, after conditioning on parental income, about half of the child income gaps remain for sons. For daughters, we find that parental income accounts for an even larger share of mobility gaps. Additional parental characteristics beyond income also explain some of the variation, but mobility gaps of 1.17 and 1.85 rank points remain for daughters and sons, respectively, even after controlling for these.

Several channels may link parental victimization to reduced intergenerational mobility. Parental income is clearly an important mediator, as the results above confirm. As documented by Bindler

[14] Results for maternal and paternal victimization are available in Appendix Table A6 for completeness.

and Ketel (2022) and Chang et al. (2025), victimization leads to persistent earnings losses, which directly constrain the resources available for investing in children's human capital. Yet the remaining mobility gaps after conditioning on income and other parental characteristics point to additional pathways. Victimization-induced depression and substance use (Papageorge et al., 2021) can affect parenting quality and reduce cognitive and non-cognitive investments during critical developmental periods (Baranov et al., 2020). The psychological and sociological mechanisms outlined in the introduction—including disrupted attachment, maladaptive coping strategies, and exposure to disorganized environments—may further limit accumulation of non-cognitive skills and social capital and in turn upward mobility. While disentangling these channels is beyond the scope of this paper, the persistence of mobility gaps even after including a rich set of parental controls suggests that the link between victimization and reduced income mobility extends beyond observable socioeconomic disadvantage.

6 Conclusion

Motivated by growing inequality[15] and, related, the challenges of fiscal sustainability, the economics literature is now densely populated with research on intergenerational correlations in outcomes. Following the lead of research on intergenerational correlations in income and education, recent studies document intergenerational correlations in crime perpetration (Besemer et al., 2017), crime incarceration (Wildeman, 2020), welfare dependence (De Haan and Schreiner, 2024), mental health (Johnston et al., 2013), and longevity (Black et al., 2023). We extend the evidence by providing estimates for the intergenerational correlation in victimization. Our finding that the intergenerational correlation is significantly attenuated among higher-income families for daughters is consistent with economic models that emphasize how parental resources shape protective investments (Becker and Tomes, 1979). Our results are consistent with income support policies helping to weaken intergenerational cycles of victimization for exposed girls. However, the weaker income gradient for sons suggests that interventions focused solely on parental resources may be insufficient to reduce boys' elevated victimization risk.

We further show that exposure to parental violence victimization is associated with significantly lower absolute income mobility—gaps of roughly 8.5 rank points for sons and 5.8 for daughters, comparable in magnitude to racial disparities documented in the United States (Chetty et al., 2020). The mobility gaps by parental violence victimization persist for sons even among higher-income families, suggesting that economic resources alone cannot fully buffer the consequences of exposure to violence.

All our estimates reflect (conditional) associations rather than purely causal effects of parental

[15] Inequality in Denmark has been rising since the 1980s (Landersø and Heckman, 2017).

victimization. Future research establishing causal links would require exogenous variation in parental victimization such as quasi-random variation in incarceration probability. Identifying mechanisms driving the intergenerational correlations, including direct exposure, income effects, disrupted parenting, or psychological trauma, could then inform targeted interventions beyond income support.

References

Aaronson, Daniel, and Bhashkar Mazumder. 2008. "Intergenerational economic mobility in the United States, 1940 to 2000." *Journal of Human Resources* 43 (1): 139–172.

Abramitzky, Ran, Leah Boustan, Elisa Jácome, and Santiago Pérez. 2021. "Intergenerational Mobility of Immigrants in the United States over Two Centuries." *American Economic Review* 111 (2): 580–608.

Adams, Abi, Kristiina Huttunen, Emily Nix, and Ning Zhang. 2024a. "The dynamics of abusive relationships." *The Quarterly Journal of Economics* 139 (4): 2135–2180.

———. 2024b. "The Economic Costs of Rape." In *2024 APPAM Fall Research Conference*.

Aizer, Anna, and Flavio Cunha. 2023. "The Production of Human Capital: Endowments, Investments, and Fertility." Edited by Orley Ashenfelter and David Card. In *Handbook of Labor Economics*, 225–348. Vols. 5A. Amsterdam: Elsevier.

Alesina, Alberto, Sebastian Hohmann, Stelios Michalopoulos, and Elias Papaioannou. 2021. "Intergenerational mobility in Africa." *Econometrica* 89 (1): 1–35.

Anderson, D. Mark, Benjamin Hansen, and Mary Beth Walker. 2013. "The minimum dropout age and student victimization." *Economics of Education Review* 35:66–74.

Asher, Sam, Paul Novosad, and Charlie Rafkin. 2024. "Intergenerational Mobility in India: New Measures and Estimates across Time and Social Groups." *American Economic Journal: Applied Economics* 16 (2): 66–98.

Bandura, Albert, and Richard H Walters. 1977. *Social learning theory*. Vol. 1. Prentice hall Englewood Cliffs, NJ.

Baranov, Victoria, Sonia Bhalotra, Pietro Biroli, and Joanna Maselko. 2020. "Maternal depression, women's empowerment, and parental investment: Evidence from a randomized controlled trial." *American economic review* 110 (3): 824–859.

Becker, Gary S, and Nigel Tomes. 1979. "An equilibrium theory of the distribution of income and intergenerational mobility." *Journal of political Economy* 87 (6): 1153–1189.

Behrman, Jere, and Paul Taubman. 1976. "Intergenerational transmission of income and wealth." *The American Economic Review* 66 (2): 436–440.

Besemer, Sytske, Shaikh I. Ahmad, Stephen P. Hinshaw, and David P. Farrington. 2017. "A systematic review and meta-analysis of the intergenerational transmission of criminal behavior." *Aggression and Violent Behavior* 37:161–178.

Bhalotra, Sonia, Damian Clarke, and Atheendar Venkataramani. 2026. "The long run economic effects of medical innovation and the role of opportunities." Forthcoming. *Journal of Political Economy*, 2026.

Bindler, Anna, Randi Hjalmarsson, Nadine Ketel, and Andreea Mitrut. 2024. "Discontinuities in the Age-Victimisation Profile and the Determinants of Victimisation." *The Economic Journal* 134 (657): 95–134.

Bindler, Anna, and Nadine Ketel. 2022. "Scaring or scarring? Labor market effects of criminal victimization." *Journal of Labor Economics* 40 (4): 939–970.

Black, Sandra E, and Paul J Devereux. 2011. "Recent developments in intergenerational mobility." *Handbook of labor economics* 4:1487–1541.

Black, Sandra E, Neil Duzett, Adriana Lleras-Muney, Nolan G Pope, and Joseph Price. 2023. *Intergenerational Transmission of Lifespan in the US*. Working Paper Series 31034.

Boustan, Leah, Mathias Fjællegaard Jensen, Ran Abramitzky, Elisa Jácome, Alan Manning, Santiago Pérez, Analysia Watley, et al. 2025. *Intergenerational Mobility of Immigrants in 15 Destination Countries*. Working Paper Series 33558.

Bowlby, John. 1979. "The bowlby-ainsworth attachment theory." *Behavioral and brain sciences* 2 (4): 637–638.

Card, David, and Gordon B. Dahl. 2011. "Family Violence and Football: The Effect of Unexpected Emotional Cues on Violent Behavior*." *The Quarterly Journal of Economics* 126 (1): 103–143.

Carneiro, Pedro, Ignacio Garcia, Kjell G. Salvanes, and Emma Tominey. 2021. "Intergenerational Mobility and the Timing of Parental Income." *Journal of Political Economy* 129 (3): 757–788.

Carneiro, Pedro, and James J. Heckman. 2003. "Human Capital Policy." Edited by James J. Heckman and Alan B. Krueger. In *Inequality in America: What Role for Human Capital Policies?* 77–239. Cambridge, MA: MIT Press.

Chalfin, Aaron, Benjamin Hansen, and Rachel Ryley. 2023. "The Minimum Legal Drinking Age and Crime Victimization." *Journal of Human Resources* 58 (4): 1141–1177.

Chang, Harrison, Shiau-Fang Chao, Kuan-Ming Chen, and Ming-Jen Lin. 2025. "Breaking Silence: How Intimate Partner Violence And Reporting Shape Later Life Outcomes." Forthcoming. *Journal of Labor Economics*, 2025.

Chetty, Raj, and Nathaniel Hendren. 2018a. "The Impacts of Neighborhoods on Intergenerational Mobility I: Childhood Exposure Effects." *Quarterly Journal of Economics* 133 (3): 1107–1162.

———. 2018b. "The Impacts of Neighborhoods on Intergenerational Mobility II: County-Level Estimates." *Quarterly Journal of Economics* 133 (3): 1163–1228.

Chetty, Raj, Nathaniel Hendren, Maggie R. Jones, and Sonya R. Porter. 2020. "Race and economic opportunity in the United States: An intergenerational perspective." *Quarterly Journal of Economics* 135 (2): 711–783.

———. 2016. "The Effects of Exposure to Better Neighborhoods on Children: New Evidence from the Moving to Opportunity Experiment." *American Economic Review* 106 (4): 855–902.

———. 2014. "Where is the land of opportunity? The geography of intergenerational mobility in the United States." *The Quarterly journal of economics* 129 (4): 1553–1623.

Collins, Matthew. 2025. "The Intergenerational Persistence of Intimate Partner Violence." Mimeo.

Cunha, Flavio, and James Heckman. 2007. "The technology of skill formation." *American economic review* 97 (2): 31–47.

Dahl, Gordon B., and Lance Lochner. 2012. "The Impact of Family Income on Child Achievement: Evidence from the Earned Income Tax Credit." *American Economic Review* 102 (5): 1927–1956.

De Haan, Monique, and Ragnhild C Schreiner. 2024. "The Intergenerational Transmission of Welfare Dependency." Mimeo.

Doyle, Joseph J, and Anna Aizer. 2018. "Economics of child protection: Maltreatment, foster care, and intimate partner violence." *Annual review of economics* 10 (1): 87–108.

Dugdale, Richard. 1877. *The Jukes: A Study of Crime, Pauperism, Disease, and Heredity*.

Dustmann, Christian, and Mikkel Mertz. 2024. "Victims of Crime." Mimeo.

Eshaghnia, Sadegh, James J Heckman, Rasmus Landersø, and Rafeh Qureshi. 2022. *Intergenerational Transmission of Family Influence*. Working Paper Series 30412.

Finlay, Keith, Michael Mueller-Smith, and Brittany Street. 2023. "Children's indirect exposure to the US justice system: Evidence from longitudinal links between survey and administrative data." *The Quarterly Journal of Economics* 138 (4): 2181–2224.

Forke, Christine M, Rachel K Myers, Joel A Fein, Marina Catallozzi, A Russell Localio, Douglas J Wiebe, and Jeane Ann Grisso. 2018. "Witnessing intimate partner violence as a child: How boys and girls model their parents' behaviors in adolescence." *Child abuse & neglect* 84:241–252.

Fortin, Nicole, Thomas Lemieux, and Sergio Firpo. 2011. "Decomposition Methods in Economics." Edited by Orley Ashenfelter and David Card. In *Handbook of Labor Economics*, arXiv: 1011.1669v3 ISSN: 15734463, 1–102. Vols. 4A. Elsevier Inc.

Hernandez-Leal, Juan D, Frances Lu, and Camila Navajas-Ahumada. 2025. "Intergenerational Persistence in Intimate-Partner Violence." Mimeo.

Hjalmarsson, Randi, Helena Holmlund, and Matthew J Lindquist. 2015. "The effect of education on criminal convictions and incarceration: Causal evidence from micro-data." *The Economic Journal* 125 (587): 1290–1326.

Hjalmarsson, Randi, and Matthew J Lindquist. 2012. "Like godfather, like son: Exploring the intergenerational nature of crime." *Journal of Human Resources* 47 (2): 550–582.

———. 2013. "The origins of intergenerational associations in crime: Lessons from Swedish adoption data." *Labour Economics* 20:68–81.

Islam, T. M., M. I. Tareque, A. D. Tiedt, and N. Hoque. 2014. "The intergenerational transmission of intimate partner violence in Bangladesh." *Global health action* 23591 (7).

Jensen, Mathias, and Alan Manning. 2025. "Background matters, but not whether parents are immigrants: outcomes of children born in Denmark." Forthcoming. *American Economic Journal: Applied Economics*, 2025.

Johnston, David W, Stefanie Schurer, and Michael A Shields. 2013. "Exploring the intergenerational persistence of mental health: Evidence from three generations." *Journal of health economics* 32 (6): 1077–1089.

Kling, Jeffrey R., Jens Ludwig, and Lawrence F. Katz. 2005. "Neighborhood Effects on Crime for Female and Male Youth: Evidence from a Randomized Housing Voucher Experiment." *Quarterly Journal of Economics* 120 (1): 87–130.

Landersø, Rasmus, and James J Heckman. 2017. "The Scandinavian fantasy: The sources of intergenerational mobility in Denmark and the US." *The Scandinavian journal of economics* 119 (1): 178–230.

Lindo, Jason M., Peter Siminski, and Isaac D. Swensen. 2018. "College Party Culture and Sexual Assault." *American Economic Journal: Applied Economics* 10 (1): 236–65.

Ludwig, Jens, Greg J. Duncan, and Paul Hirschfield. 2001. "Urban Poverty and Juvenile Crime: Evidence from a Randomized Housing-Mobility Experiment." *Quarterly Journal of Economics* 116 (2): 655–679.

Madruga, Clarice S., Maria Carmen Viana, Renata Rigacci Abdalla, Raul Caetano, and Ronaldo Laranjeira. 2017. "Pathways from witnessing parental violence during childhood to involvement in intimate partner violence in adult life: The roles of depression and substance use." *Drug and Alcohol Review* 36 (1): 107–114.

Papageorge, Nicholas W, Gwyn C Pauley, Mardge Cohen, Tracey E Wilson, Barton H Hamilton, and Robert A Pollak. 2021. "Health, human capital, and domestic violence." *Journal of Human Resources* 56 (4): 997–1030.

Pollak, Robert A. 2004. "An Intergenerational Model of Domestic Violence." *Journal of Population Economics* 17 (2): 311–329.

Puno, Abigail, Ranyeong Kim, Joshua Jeong, Jinho Kim, and Rockli Kim. 2023. "Intergenerational transmission of intimate partner violence among women: Evidence from the 2017 Philippines National Demographic and Health Survey." *SSM - Population Health* 23:101392.

Rees, Daniel I., and Kevin T. Schnepel. 2009. "College Football Games and Crime." *Journal of Sports Economics* 10 (1): 68–87.

Sampson, Robert J, and John H Laub. 1997. "A life-course theory of cumulative disadvantage and the stability of delinquency." *Developmental theories of crime and delinquency* 7:133–161.

Shaw, Clifford R, and Henry D McKay. 1942. *Juvenile delinquency and urban areas.* University of Chicago press.

Van de Weijer, Steve GA, Catrien CJH Bijleveld, and Arjan AJ Blokland. 2014. "The intergenerational transmission of violent offending." *Journal of family violence* 29 (2): 109–118.

Van der Kolk, Bessel A. 1989. "The compulsion to repeat the trauma: Re-enactment, revictimization, and masochism." *Psychiatric Clinics of North America* 12 (2): 389–411.

Wildeman, Christopher. 2020. "The intergenerational transmission of criminal justice contact." *Annual Review of Criminology* 3 (1): 217–244.

Wood, Samantha L., and Marilyn S. Sommers. 2011. "Consequences of Intimate Partner Violence on Child Witnesses: A Systematic Review of the Literature." *Journal of Child and Adolescent Psychiatric Nursing* 24 (4): 223–236. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1744-6171.2011.00302.x.

Intergenerational Transmission of Victimization

Online Appendix

Sonia Bhalotra
University of Warwick

N. Meltem Daysal
University of Copenhagen, CEBI, CESifo, and IZA

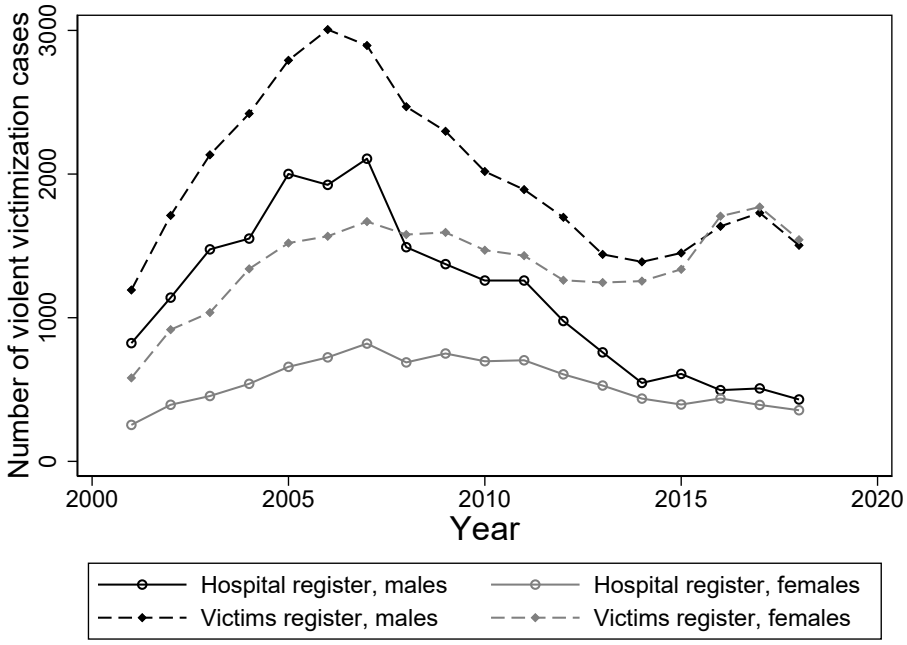
Mathias Fjællegaard Jensen
University of Oxford

Thomas Jørgensen
University of Copenhagen

Sébastien Montpetit
University of Warwick

Appendix A

Transmission of Victimization: Additional Results



Appendix Figure A1: Violent crime victimization in hospitalization and victimization registers, 2001-2018

Note: This Figure plots the time series of violent crime victimization measured in the Danish hospitalization and victimization registers, separately for men and women. Victimization outcomes are measured at ages 18-35 for the 1980-1987 birth cohorts.

Appendix Table A1: Intergenerational transmission of assault victimization, sons

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Any parent						
Parents$_i$	11.53***	11.80***	10.61***	7.48***	5.39***	3.77***
	(0.57)	(0.57)	(0.57)	(0.57)	(0.58)	(0.59)
Mean non-exposed	13.45	13.45	13.45	13.45	13.45	13.45
N	188,010	188,010	188,010	188,010	188,010	188,010
Panel B: Fathers						
Father$_i$	10.69***	10.96***	9.72***	6.64***	4.49***	2.79***
	(0.71)	(0.71)	(0.70)	(0.71)	(0.71)	(0.72)
Mean non-exposed	13.60	13.60	13.60	13.60	13.60	13.60
N	188,010	188,010	188,010	188,010	188,010	188,010
Panel C: Mothers						
Mother$_i$	13.56***	13.79***	12.63***	9.14***	6.94***	5.70***
	(0.90)	(0.90)	(0.90)	(0.90)	(0.90)	(0.91)
Mean non-exposed	13.63	13.63	13.63	13.63	13.63	13.63
N	188,010	188,010	188,010	188,010	188,010	188,010
Cohort FEs	No	Yes	Yes	Yes	Yes	Yes
Municipality FEs	No	No	Yes	Yes	Yes	Yes
Parents' cohab.	No	No	No	Yes	Yes	Yes
SES controls	No	No	No	No	Yes	Yes
Other criminal status	No	No	No	No	No	Yes

Notes: This table reports regression results of equations (1) for sons. Victimization outcomes are observed at child age 8-17 for parents, and child age 18-35 for children. The dependent variable is whether the child has an assault-related emergency-room hospital contact between ages 18 and 35. Income ranks (0-100) are determined within child birth-year cohorts, regardless of victimization status. Point estimates (and standard errors) are rescaled by 100 to ease interpretation and readability. Robust standard errors in parentheses. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table A2: Intergenerational transmission of assault victimization, daughters

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Any parent						
Parents$_i$	8.94***	8.96***	8.43***	6.24***	4.49***	3.59***
	(0.45)	(0.45)	(0.45)	(0.45)	(0.44)	(0.45)
Mean non-exposed	4.27	4.27	4.27	4.27	4.27	4.27
N	182,031	182,031	182,031	182,031	182,031	182,031
Panel B: Fathers						
Father$_i$	7.70***	7.71***	7.15***	4.96***	3.21***	2.27***
	(0.53)	(0.53)	(0.53)	(0.53)	(0.53)	(0.53)
Mean non-exposed	4.40	4.40	4.40	4.40	4.40	4.40
N	182,031	182,031	182,031	182,031	182,031	182,031
Panel C: Mothers						
Mother$_i$	11.53***	11.54***	10.96***	8.50***	6.57***	5.80***
	(0.76)	(0.76)	(0.76)	(0.76)	(0.75)	(0.76)
Mean non-exposed	4.41	4.41	4.41	4.41	4.41	4.41
N	182,031	182,031	182,031	182,031	182,031	182,031
Cohort FEs	No	Yes	Yes	Yes	Yes	Yes
Municipality FEs	No	No	Yes	Yes	Yes	Yes
Parents' cohab.	No	No	No	Yes	Yes	Yes
SES controls	No	No	No	No	Yes	Yes
Other criminal status	No	No	No	No	No	Yes

Notes: This table reports regression results of equations (1) for daughters. Victimization outcomes are observed at child age 8-17 for parents, and child age 18-35 for children. The dependent variable is whether the child has an assault-related emergency-room hospital contact between ages 18 and 35. Income ranks (0-100) are determined within child birth-year cohorts, regardless of victimization status. Point estimates (and standard errors) are rescaled by 100 to ease interpretation and readability. Robust standard errors in parentheses. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table A3: Oaxaca-Blinder decompositions of child victimization between children of victimized and non-victimized parents for sons

	(1)	(2)	(3)	(4)	(5)
Panel A: Any parent					
Victimized	24.98***	24.98***	24.98***	24.98***	24.98***
parent	(0.561)	(0.137)	(0.145)	(0.167)	(0.169)
Non-victimized	13.45***	13.45***	13.45***	13.45***	13.45***
parent	(0.0800)	(0.00871)	(0.0108)	(0.0123)	(0.0127)
Difference	11.53***	11.53***	11.53***	11.53***	11.53***
	(0.567)	(0.137)	(0.146)	(0.167)	(0.169)
Total explained	-0.264***	0.923***	4.056***	6.146***	7.152***
	(0.0289)	(0.0456)	(0.0609)	(0.0787)	(0.0874)
Total unexplained	11.80***	10.61***	7.477***	5.386***	4.381***
	(0.568)	(0.122)	(0.125)	(0.137)	(0.136)
Panel B: Fathers					
Victimized	24.29***	24.29***	24.29***	24.29***	24.29***
parent	(0.702)	(0.198)	(0.207)	(0.238)	(0.240)
Non-victimized	13.60***	13.60***	13.60***	13.60***	13.60***
parent	(0.0799)	(0.00874)	(0.0110)	(0.0126)	(0.0129)
Difference	10.69***	10.69***	10.69***	10.69***	10.69***
	(0.707)	(0.198)	(0.208)	(0.238)	(0.240)
Total explained	-0.265***	0.968***	4.049***	6.198***	7.212***
	(0.0315)	(0.0590)	(0.0794)	(0.103)	(0.113)
Total unexplained	10.96***	9.722***	6.641***	4.493***	3.478***
	(0.707)	(0.180)	(0.187)	(0.205)	(0.206)
Panel C: Mothers					
Victimized	27.19***	27.19***	27.19***	27.19***	27.19***
parent	(0.896)	(0.317)	(0.331)	(0.380)	(0.382)
Non-victimized	13.63***	13.63***	13.63***	13.63***	13.63***
parent	(0.0797)	(0.00878)	(0.0110)	(0.0126)	(0.0128)
Difference	13.56***	13.56***	13.56***	13.56***	13.56***
	(0.900)	(0.317)	(0.332)	(0.381)	(0.382)
Total explained	-0.229***	0.925***	4.422***	6.617***	7.627***
	(0.0298)	(0.0682)	(0.0935)	(0.123)	(0.140)
Total unexplained	13.79***	12.63***	9.136***	6.941***	5.931***
	(0.900)	(0.305)	(0.311)	(0.352)	(0.354)
Cohort FEs	Yes	Yes	Yes	Yes	Yes
Municipality FEs	No	Yes	Yes	Yes	Yes
Parents' cohab.	No	No	Yes	Yes	Yes
SES controls	No	No	No	Yes	Yes
Parental criminal status	No	No	No	No	Yes

Notes: This Table reports results of Oaxaca-Blinder decompositions of differences in victimization rates between children who are exposed to victimization and those who are not. Coefficients of children without any victimized parent used as reference level. Victimization outcomes are observed at child age 8-17 for parents, and child age 18-35 for the children. Point estimates (and standard errors) are rescaled by 100 to ease interpretation and readability. Robust standard errors in parentheses. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table A4: Oaxaca-Blinder decompositions of child victimization between children of victimized and non-victimized parents for daughters

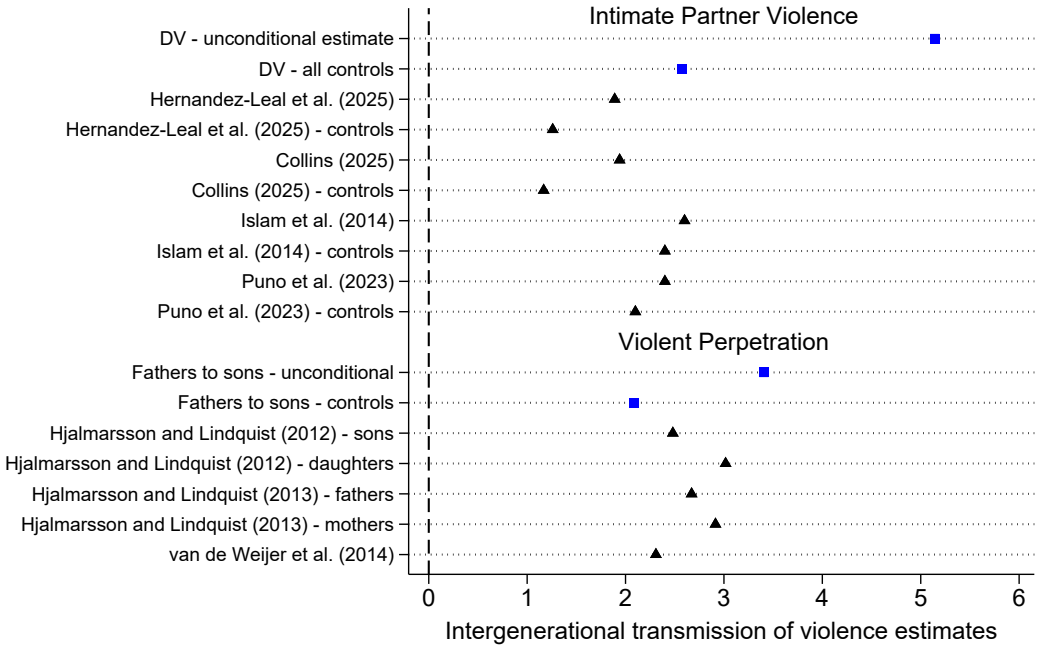
	(1)	(2)	(3)	(4)	(5)
Panel A: Any parent					
Victimized parent	13.21***	13.21***	13.21***	13.21***	13.21***
	(0.444)	(0.0919)	(0.107)	(0.135)	(0.136)
Non-victimized parents	4.273***	4.273***	4.273***	4.273***	4.273***
	(0.0482)	(0.00387)	(0.00576)	(0.00731)	(0.00746)
Difference	8.939***	8.939***	8.939***	8.939***	8.939***
	(0.447)	(0.0919)	(0.107)	(0.135)	(0.136)
Total explained	-0.0253	0.510***	2.699***	4.445***	5.018***
	(0.0180)	(0.0210)	(0.0358)	(0.0550)	(0.0600)
Total unexplained	8.965***	8.429***	6.240***	4.494***	3.921***
	(0.447)	(0.0845)	(0.0889)	(0.104)	(0.103)
Panel B: Fathers					
Victimized parent	12.10***	12.10***	12.10***	12.10***	12.10***
	(0.531)	(0.130)	(0.146)	(0.182)	(0.183)
Non-victimized parents	4.398***	4.398***	4.398***	4.398***	4.398***
	(0.0486)	(0.00397)	(0.00611)	(0.00775)	(0.00788)
Difference	7.698***	7.698***	7.698***	7.698***	7.698***
	(0.533)	(0.130)	(0.146)	(0.182)	(0.183)
Total explained	-0.0105	0.545***	2.735***	4.492***	5.085***
	(0.0187)	(0.0268)	(0.0467)	(0.0715)	(0.0779)
Total unexplained	7.709***	7.154***	4.963***	3.206***	2.613***
	(0.533)	(0.122)	(0.128)	(0.150)	(0.151)
Panel C: Mothers					
Victimized	15.94***	15.94***	15.94***	15.94***	15.94***
parent	(0.762)	(0.762)	(0.268)	(0.314)	(0.315)
Non-victimized	4.412***	4.412***	4.412***	4.412***	4.412***
parent	(0.0484)	(0.0484)	(0.00605)	(0.00771)	(0.00777)
Difference	11.53***	11.53***	11.53***	11.53***	11.53***
	(0.764)	(0.764)	(0.268)	(0.314)	(0.315)
Total explained	-0.00829	0.570***	3.037***	4.958***	5.523***
	(0.0184)	(0.0454)	(0.0571)	(0.0882)	(0.0962)
Total unexplained	11.54***	10.96***	8.495***	6.575***	6.009***
	(0.764)	(0.761)	(0.245)	(0.278)	(0.275)
Cohort FEs	Yes	Yes	Yes	Yes	Yes
Municipality FEs	No	Yes	Yes	Yes	Yes
Parents' cohab.	No	No	Yes	Yes	Yes
SES controls	No	No	No	Yes	Yes
Parental criminal status	No	No	No	No	Yes

Notes: This Table reports results of Oaxaca-Blinder decompositions of differences in victimization rates between children who are exposed to victimization and those who are not. Coefficients of children without any victimized parent used as reference level. Victimization outcomes are observed at child age 8-17 for parents, and child age 18-35 for the children. Point estimates (and standard errors) are rescaled by 100 to ease interpretation and readability. Robust standard errors in parentheses. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table A5: Intergenerational transmission of assault victimization, heterogeneity by cumulative exposure

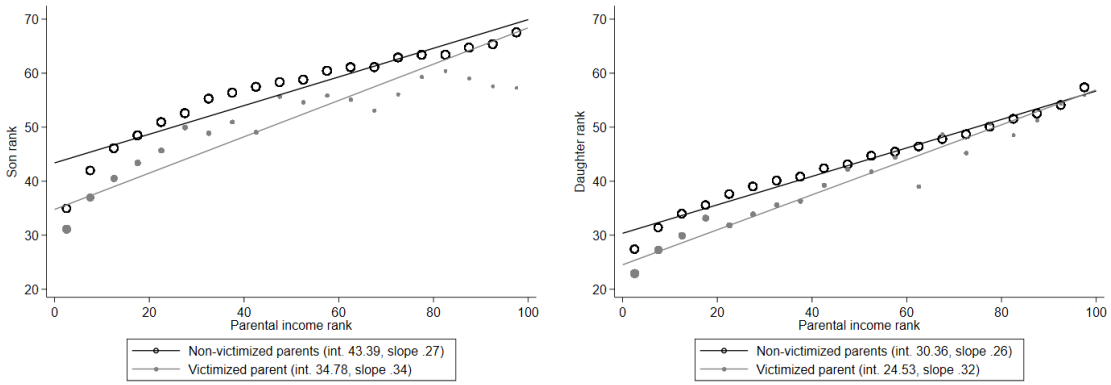
Dep. var:	Sons		Daughters	
	Father	Mother	Father	Mother
Middle childhood$_i$ (8-12 years old)	3.01*	3.45*	1.48	7.08***
	(1.22)	(1.51)	(0.87)	(1.43)
Middle and adolescence$_i$ (8-12 & 13-17 y.o.)	3.08	10.40**	8.36**	2.83
	(2.78)	(3.35)	(2.70)	(2.68)
All controls	Yes	Yes	Yes	Yes
Observations	188,010	188,010	182,031	182,031
R-squared	0.028	0.028	0.030	0.030

Notes: This Table reports regression results of the following model: Child violence$_i$ = γ_1+γ_2 Middle childhood$_i$+γ_3 Middle and adolescence$_i$+ $X'\delta + \eta_i$, where the exposure measures take the value of one if at least one of child i's parents was victimized when the child was aged 8-12, and victimized in both middle childhood (8-12 years old) and youth (13-17), respectively. Children outcomes are measured at child age 18-35. Point estimates (and standard errors) are rescaled by 100 to ease interpretation and readability. Robust standard errors in parentheses. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.



Appendix Figure A2: Comparison of intergenerational transmission estimates with previous literature

Note: Estimates are reported as odds ratios. For studies (including ours) that report linear-probability estimates, we convert the reported estimates to odds ratios using the corresponding baseline risks. The proxy for domestic violence (DV) for the reported estimates in blue (squares) is assault victimization occurring in a residential area. Estimates from previous studies are displayed as black triangles.



(a) Assault victimization, sons

(b) Assault victimization, daughters

Appendix Figure A3: Rank-rank relationships by parental victimization and child gender

Note: These figures display the non-linear rank-rank relationships along with linear fits, separately for sons and daughters. Children of victims are in light gray and children of non-victimized parents are in black. Each data point represents the average income rank of a child in a given ventile of the parental income distribution. Victimization outcomes are observed at child age 8-17 for parents, and child age 18-35 for the children. The dependent variable is the child's income rank where its income is averaged over ages 30-35. Income ranks, 0-100, are determined within child birth-year cohorts, regardless of crime status. Symbol sizes are weighted by the number of observations.

Appendix Table A6: Social mobility by parental violence victimization and child gender

Victimized parent:	Father				Mother			
Child gender:	Daughters		Sons		Daughters		Sons	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Victimized parent$_i$ (α_2)	-5.512***	-0.824	-6.791***	0.507	-8.819***	-2.117***	-13.39***	-3.839***
	(0.592)	(0.586)	(0.744)	(0.744)	(0.705)	(0.700)	(0.873)	(0.873)
Parents' rank$_i$×	0.044***	0.009	0.033*	-0.045**	0.061***	0.012	0.090***	0.010
Victimized parent$_i$ (β_2)	(0.016)	(0.016)	(0.020)	(0.019)	(0.020)	(0.020)	(0.023)	(0.022)
Parents' rank$_i$ (β_1)	0.218***	0.116***	0.228***	0.135***	0.215***	0.086***	0.181***	0.062***
	(0.002)	(0.003)	(0.003)	(0.004)	(0.002)	(0.003)	(0.004)	(0.004)
Constant (α_1)	32.77***	28.57***	45.62***	35.37***	33.05***	29.09***	48.25***	37.13***
	(0.129)	(5.756)	(0.150)	(10.10)	(0.126)	(5.573)	(0.148)	(9.784)
Parental controls	No	Yes	No	Yes	No	Yes	No	Yes
Observations	182,031	182,031	188,010	188,010	182,031	182,031	188,010	188,010
R-squared	0.056	0.107	0.048	0.086	0.057	0.103	0.033	0.079

Notes: This Table reports results of rank-rank regressions allowing for heterogeneous effects of parental income by parental violence victimization following equation (2). Violence outcomes are observed at child age 8-17 for parents, and child age 18-35 for the children. The dependent variable is the child's income rank where its income is averaged over ages 30-35. Income ranks, 0-100, are determined within child birth-year cohorts, regardless of crime status. Robust standard errors in parentheses. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table A7: Oaxaca-Blinder decompositions of social mobility gaps between children exposed to victimization and children not exposed

	Daughters		Sons	
	(1)	(2)	(3)	(4)
Child income rank				
Exposed to violence	33.76***	33.76***	44.86***	44.86***
	(0.33)	(0.33)	(0.40)	(0.40)
Not exposed	44.49***	44.49***	57.90***	57.90***
	(0.06)	(0.06)	(0.07)	(0.07)
Difference	-10.74***	-10.74***	-13.05***	-13.05***
	(0.34)	(0.34)	(0.41)	(0.41)
Explained share of mobility gaps				
Explained gap	-6.83***	-9.57***	-6.55***	-11.19***
	(0.11)	(0.15)	(0.11)	(0.17)
Unexplained gap	-3.90***	-1.17***	-6.50***	-1.85***
	(0.32)	(0.33)	(0.40)	(0.40)
Observations	182,031	182,031	188,010	188,010

Notes: This Table reports results of Oaxaca-Blinder decompositions of differences in child income rank between children of victimized parents to those not exposed. Coefficients of children without a violent victimized parent used as reference level. In odd columns, only parental income rank is used as regressor. Regressions in even columns include the full set of covariates. Victimization outcomes are observed at child age 8-17 for parents, and child age 18-35 for the children. The dependent variable is the child's income rank where its income is averaged over ages 30-35. Income ranks, 0-100, are determined within child birth-year cohorts, regardless of victimization status. Robust standard errors in parentheses. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix B

Domestic Violence

Appendix Table B1: Intergenerational transmission of domestic violence victimization, sons

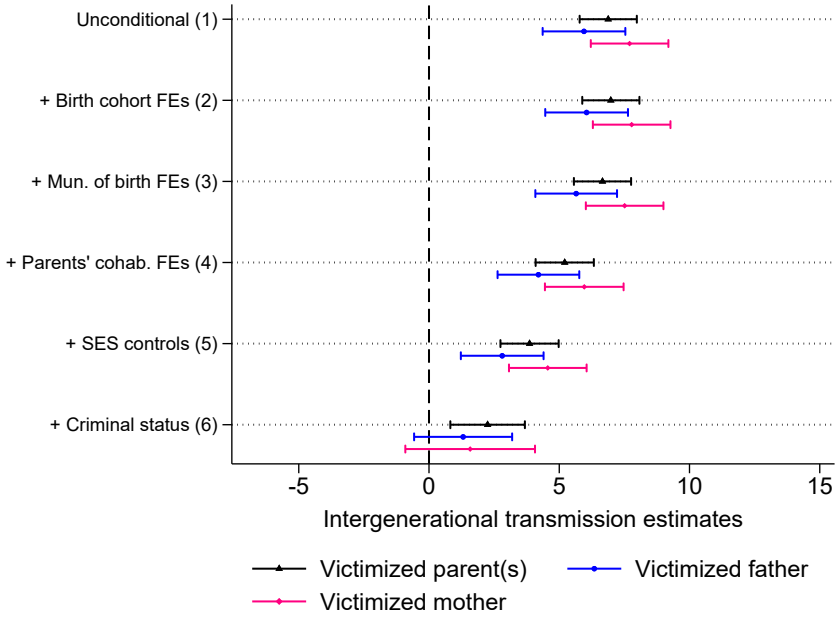
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Any parent						
Parents$_i$ (γ_2)	6.88***	6.98***	6.66***	5.21***	3.86***	2.25***
	(0.56)	(0.56)	(0.56)	(0.57)	(0.57)	(0.73)
Mean non-exposed (γ_1)	3.49	3.49	3.49	3.49	3.49	3.49
N	188,010	188,010	188,010	188,010	188,010	188,010
R^2	0.002	0.002	0.008	0.012	0.018	0.019
Panel B: Fathers						
Father$_i$	5.95***	6.05***	5.65***	4.20***	2.81***	1.31
	(0.81)	(0.81)	(0.80)	(0.80)	(0.81)	(0.96)
Mean non-exposed (γ_1)	3.56	3.56	3.56	3.56	3.56	3.56
N	188,010	188,010	188,010	188,010	188,010	188,010
R^2	0.001	0.001	0.007	0.011	0.017	0.018
Panel C: Mothers						
Mother$_i$	7.70***	7.78***	7.51***	5.96***	4.56***	1.58
	(0.76)	(0.76)	(0.76)	(0.77)	(0.76)	(1.27)
Mean non-exposed (γ_1)	3.53	3.53	3.53	3.53	3.53	3.53
N	188,010	188,010	188,010	188,010	188,010	188,010
R^2	0.002	0.002	0.008	0.012	0.018	0.018
Cohort FEs	No	Yes	Yes	Yes	Yes	Yes
Municipality FEs	No	No	Yes	Yes	Yes	Yes
Parents' cohab.	No	No	No	Yes	Yes	Yes
SES controls	No	No	No	No	Yes	Yes
Other criminal status	No	No	No	No	No	Yes

Notes: This table reports regression results of equations (1) for sons. Victimization outcomes are observed at child age 8-17 for parents, and child age 18-35 for children. The dependent variable is whether the child has an assault-related emergency-room hospital contact recorded as occurring in a residential area between ages 18 and 35. Income ranks (0-100) are determined within child birth-year cohorts, regardless of victimization status. Point estimates (and standard errors) are rescaled by 100 to ease interpretation and readability. Robust standard errors in parentheses. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

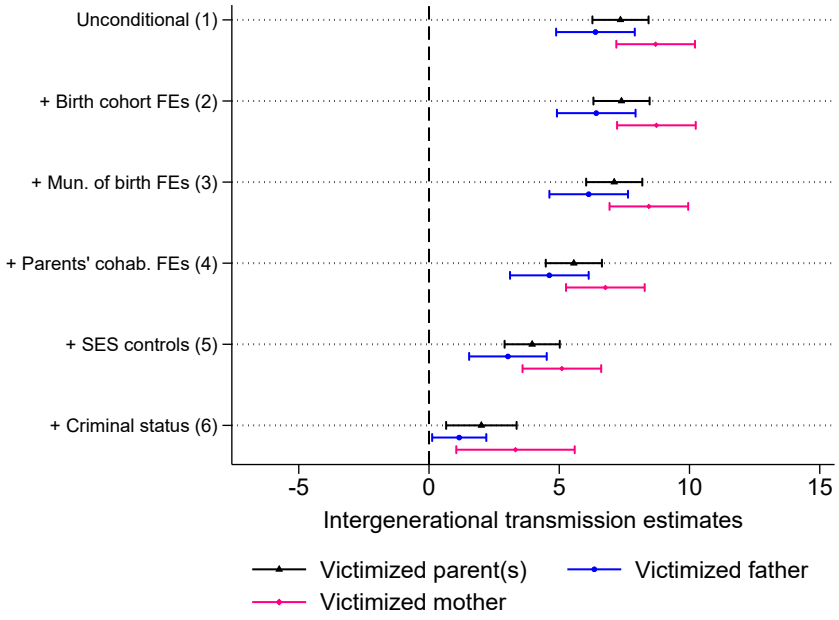
Appendix Table B2: Intergenerational transmission of domestic violence victimization, daughters

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Any parent						
Parents$_i$ (γ_2)	7.35***	7.39***	7.11***	5.56***	3.96***	2.01***
	(0.55)	(0.55)	(0.55)	(0.55)	(0.54)	(0.69)
Mean non-exposed (γ_1)	2.06	2.06	2.06	2.06	2.06	2.06
N	182,031	182,031	182,031	182,031	182,031	182,031
R^2	0.004	0.004	0.008	0.015	0.025	0.026
Panel B: Fathers						
Father$_i$	6.39***	6.42***	6.13***	4.62***	3.03***	1.16
	(0.77)	(0.77)	(0.77)	(0.77)	(0.76)	(0.53)
Mean non-exposed (γ_1)	2.13	2.13	2.13	2.13	2.13	2.13
N	182,031	182,031	182,031	182,031	182,031	182,031
R^2	0.001	0.001	0.006	0.013	0.024	0.025
Panel C: Mothers						
Mother$_i$	8.70***	8.73***	8.44***	6.77***	5.10***	3.32***
	(0.77)	(0.77)	(0.77)	(0.77)	(0.77)	(1.16)
Mean non-exposed (γ_1)	2.10	2.10	2.10	2.10	2.10	2.10
N	182,031	182,031	182,031	182,031	182,031	182,031
R^2	0.003	0.003	0.007	0.015	0.025	0.025
Cohort FEs	No	Yes	Yes	Yes	Yes	Yes
Municipality FEs	No	No	Yes	Yes	Yes	Yes
Parents' cohab.	No	No	No	Yes	Yes	Yes
SES controls	No	No	No	No	Yes	Yes
Other criminal status	No	No	No	No	No	Yes

Notes: This table reports regression results of equations (1) for daughters. Victimization outcomes are observed at child age 8-17 for parents, and child age 18-35 for children. The dependent variable is whether the child has an assault-related emergency-room hospital contact recorded as occurring in a residential area between ages 18 and 35. Income ranks (0-100) are determined within child birth-year cohorts, regardless of victimization status. Point estimates (and standard errors) are rescaled by 100 to ease interpretation and readability. Robust standard errors in parentheses. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.



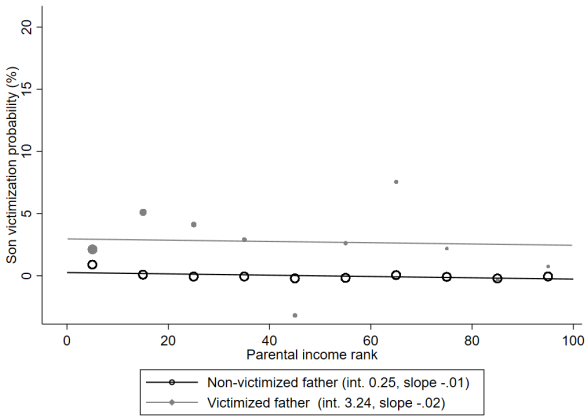
(a) Domestic violence, sons



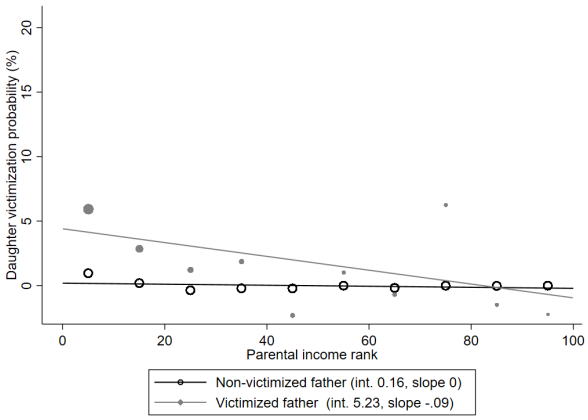
(b) Domestic violence, daughters

Appendix Figure B1: Intergenerational transmission of domestic violence victimization

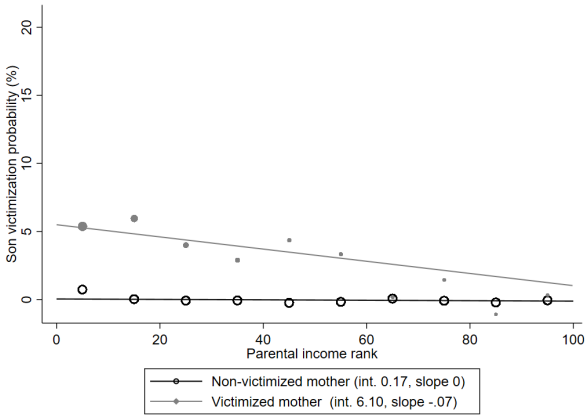
Note: These figures report estimates of domestic violence transmission (γ_2) using equation (1). Victimization outcomes are observed at child age 8-17 for parents, and child age 18-35 for children. The dependent variable is whether the child has an assault-related emergency-room hospital contact recorded as occurring in a residential area between ages 18 and 35. Income ranks (0-100) are determined within child birth-year cohorts, regardless of victimization status. Point estimates (and standard errors) are rescaled by 100 to ease interpretation and readability. Robust standard errors in parentheses. 95% confidence intervals are reported.



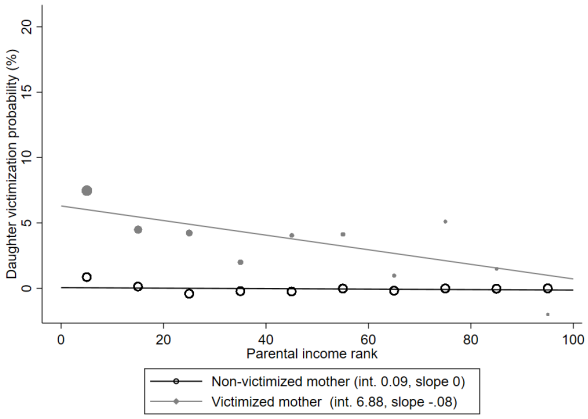
(a) Fathers to Sons



(b) Fathers to Daughters



(c) Mothers to Sons



(d) Mothers to Daughters

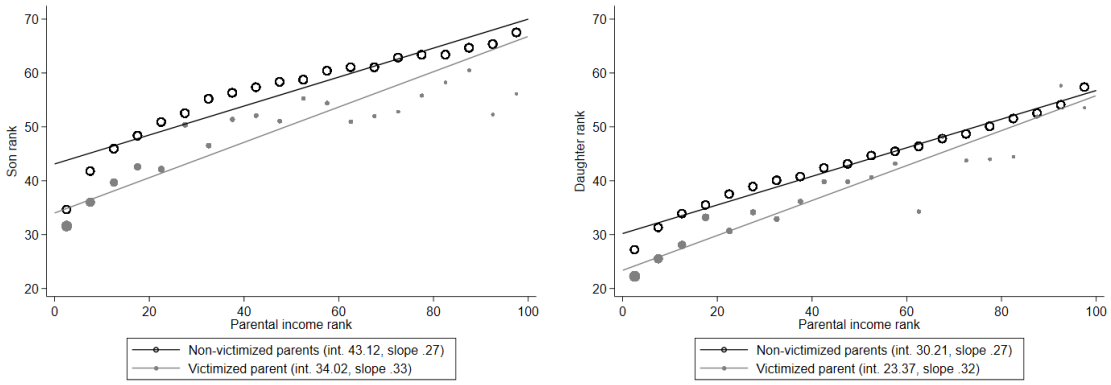
Appendix Figure B2: Child domestic violence victimization outcomes across parental income deciles by parental domestic violence victimization

Note: Each data point represents the domestic violence rates at a given decile of the parental income distribution residualized with respect to all control variables except parental income. Violence outcomes are observed at child age 8-17 for parents, and child age 18-35 for the children. The dependent variable is whether the child appears in the emergency-room hospitalization data for an assault in residential areas between the ages of 18 to 35. Income ranks, 0-100, are determined within child birth-year cohorts, regardless of victimization status. Cells with low counts are not reported for confidentiality reasons. Symbol sizes are weighted by the number of observations.

Appendix Table B3: Social mobility by parental domestic violence victimization and child gender

Dep. var: child's income rank				
	Daughters		Sons	
	(1)	(2)	(3)	(4)
Victimized parent$_i$ (α_2)	-6.690***	-2.470***	-9.203***	-2.195***
	(0.621)	(0.626)	(0.790)	(0.802)
Parents' rank$_i$×	0.057***	0.024	0.061***	-0.012
Victimized parent$_i$ (β_2)	(0.021)	(0.021)	(0.024)	(0.024)
Parents' rank$_i$ (β_1)	0.269***	0.151***	0.259***	0.155***
	(0.002)	(0.003)	(0.002)	(0.004)
Constant (α_1)	29.99***	25.56***	43.95***	33.86***
	(0.126)	(5.574)	(0.149)	(10.10)
Parental controls	No	Yes	No	Yes
Observations	182,031	182,031	188,010	188,010
R-squared	0.086	0.110	0.062	0.087

Notes: This Table reports results of rank-rank regressions allowing for heterogeneous effects of parental income by parental domestic violence victimization following equation (2). Victimization outcomes are observed at child age 8-17 for parents. The dependent variable is the child's income rank where its income is averaged over ages 30-35. Income ranks, 0-100, are determined within child birth-year cohorts, regardless of victimization status. Point estimates (and standard errors) are rescaled by 100 to ease interpretation and readability. Robust standard errors in parentheses. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.



(a) Domestic violence, sons (b) Domestic violence, daughters

Appendix Figure B3: Rank-rank relationships by parental domestic violence victimization and child gender

Note: These figures display the non-linear rank-rank relationships along with linear fits, separately for sons and daughters. Children of victims are in light gray and children of non-victimized parents are in black. Each data point represents the average income rank of a child in a given ventile of the parental income distribution. Victimization outcomes are observed at child age 8-17 for parents, and child age 18-35 for the children. The dependent variable is the child's income rank where its income is averaged over ages 30-35. Income ranks, 0-100, are determined within child birth-year cohorts, regardless of crime status. Symbol sizes are weighted by the number of observations.

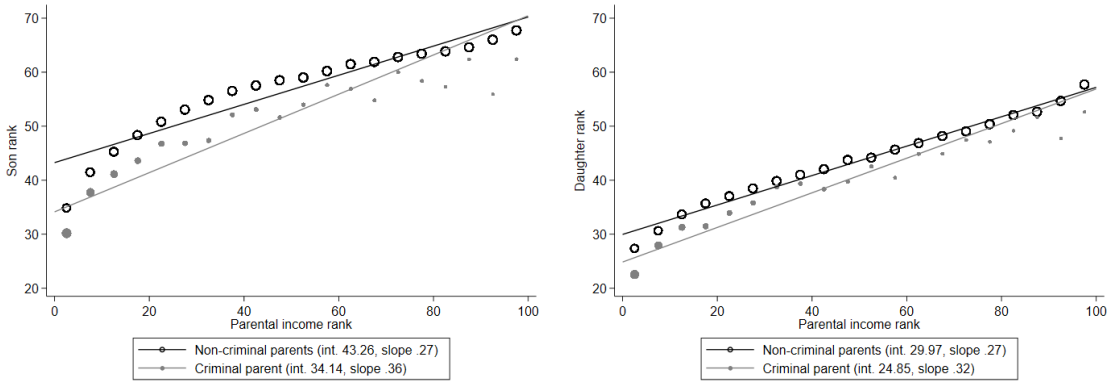
Appendix C

Violent Perpetration

Appendix Table C1: Descriptive Statistics, violent perpetration

		Child violence by parental violence		Parental violence		
	Full sample	Exposed	Non-exp.	Father	Mother	Exposed children
All children	4.99	17.15	4.67	2.34	0.26	9,431
	(21.77)	(37.69)	(21.10)	(15.13)	(5.13)	
Sons	8.67	28.47	8.16	2.31	0.26	4,583
	(28.14)	(45.13)	(27.37)	(15.02)	(5.11)	
Daughters	1.19	5.73	1.07	2.38	0.27	4,851
	(10.82)	(23.24)	(10.27)	(15.24)	(5.15)	

Notes: The data source is the Danish population (BEF) and crime registers (KRSI). Perpetration outcomes are observed at child age 8-17 for parents, and child age 18-35 for children. Statistics are rescaled by 100 to reflect percentages. The last column reports the number of children exposed to at least one violent parent in a given exposure-child gender cell. The number of observations is 370,041 for the full sample; 188,010 for sons; and 182,031 for daughters.



(a) Violence perpetration, sons

(b) Violence perpetration, daughters

Appendix Figure C1: Rank-rank relationships by parental violent perpetration and child gender

Note: These figures display the non-linear rank-rank relationships along with linear fits, separately for sons and daughters. Children of perpetrators are in light gray and children of non-criminal parents are in black. Each data point represents the average income rank of a child in a given ventile of the parental income distribution. Violent perpetration outcomes are observed at child age 8-17 for parents, and child age 18-35 for the children. The dependent variable is the child's income rank where its income is averaged over ages 30-35. Income ranks, 0-100, are determined within child birth-year cohorts, regardless of crime status. Symbol sizes are weighted by the number of observations.

Appendix Table C2: Intergenerational transmission of violent perpetration, sons

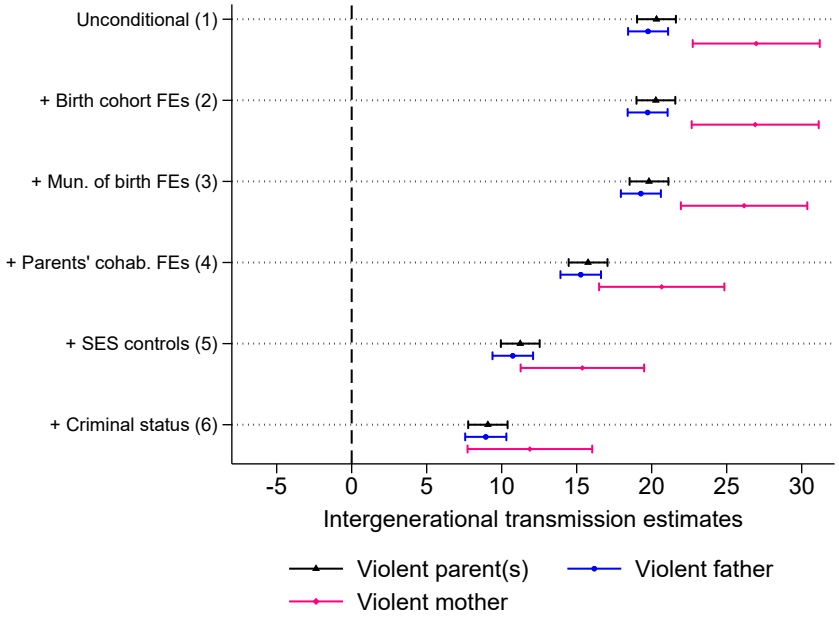
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Any parent						
Parents$_i$ (γ_2)	20.32***	20.28***	19.82***	15.76***	11.24***	9.08***
	(0.66)	(0.66)	(0.66)	(0.66)	(0.66)	(0.67)
Mean non-exposed (γ_1)	8.16	8.16	8.16	8.16	8.16	8.16
N	188,010	188,010	188,010	188,010	188,010	188,010
R^2	0.013	0.013	0.018	0.032	0.054	0.057
Panel B: Fathers						
Father$_i$	19.76***	19.73***	19.28***	15.27***	10.74***	8.94***
	(0.68)	(0.68)	(0.68)	(0.69)	(0.69)	(0.70)
Mean non-exposed (γ_1)	8.21	8.21	8.21	8.21	8.21	8.21
N	188,010	188,010	188,010	188,010	188,010	188,010
R^2	0.011	0.011	0.016	0.031	0.053	0.055
Panel C: Mothers						
Mother$_i$	26.97***	26.90***	26.16***	20.67***	15.38***	11.88***
	(2.16)	(2.16)	(2.15)	(2.13)	(2.10)	(2.12)
Mean non-exposed (γ_1)	8.60	8.60	8.60	8.60	8.60	8.60
N	188,010	188,010	188,010	188,010	188,010	188,010
R^2	0.002	0.003	0.008	0.026	0.051	0.053
Cohort FEs	No	Yes	Yes	Yes	Yes	Yes
Municipality FEs	No	No	Yes	Yes	Yes	Yes
Parents' cohab.	No	No	No	Yes	Yes	Yes
SES controls	No	No	No	No	Yes	Yes
Other criminal status	No	No	No	No	No	Yes

Notes: This table reports regression results of equations (1) for sons. Violence outcomes are observed at child age 8-17 for parents, and child age 18-35 for children. The dependent variable is whether the child appears in the crime register for violence perpetration between ages 18 and 35. Income ranks (0-100) are determined within child birth-year cohorts, regardless of crime status. Point estimates (and standard errors) are rescaled by 100 to ease interpretation and readability. Robust standard errors in parentheses. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

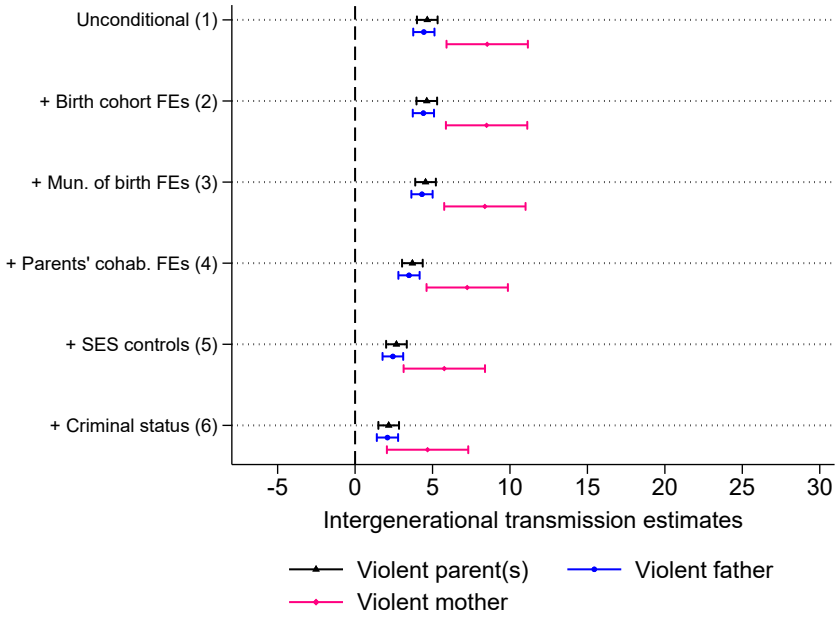
Appendix Table C3: Intergenerational transmission of violent perpetration, daughters

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Any parent						
Parents$_i$ (γ_2)	4.66***	4.63***	4.55***	3.70***	2.67***	2.17***
	(0.34)	(0.34)	(0.34)	(0.34)	(0.34)	(0.34)
Mean non-exposed (γ_1)	1.07	1.07	1.07	1.07	1.07	1.07
N	182,031	182,031	182,031	182,031	182,031	182,031
R^2	0.005	0.005	0.007	0.011	0.018	0.019
Panel B: Fathers						
Father$_i$	4.44***	4.41***	4.32***	3.48***	2.44***	2.09***
	(0.35)	(0.35)	(0.35)	(0.35)	(0.34)	(0.35)
Mean non-exposed (γ_1)	1.08	1.08	1.08	1.08	1.08	1.08
N	182,031	182,031	182,031	182,031	182,031	182,031
R^2	0.004	0.004	0.006	0.011	0.018	0.018
Panel C: Mothers						
Mother$_i$	8.53***	8.49***	8.38***	7.24***	5.76***	4.68***
	(1.34)	(1.34)	(1.34)	(1.34)	(1.34)	(1.34)
Mean non-exposed (γ_1)	1.16	1.16	1.16	1.16	1.16	1.16
N	182,031	182,031	182,031	182,031	182,031	182,031
R^2	0.002	0.002	0.004	0.010	0.017	0.018
Cohort FEs	No	Yes	Yes	Yes	Yes	Yes
Municipality FEs	No	No	Yes	Yes	Yes	Yes
Parents' cohab.	No	No	No	Yes	Yes	Yes
SES controls	No	No	No	No	Yes	Yes
Other criminal status	No	No	No	No	No	Yes

Notes: This table reports regression results of equations (1) for daughters. Violence outcomes are observed at child age 8-17 for parents, and child age 18-35 for children. The dependent variable is whether the child appears in the crime register for violence perpetration between ages 18 and 35. Income ranks (0-100) are determined within child birth-year cohorts, regardless of crime status. Point estimates (and standard errors) are rescaled by 100 to ease interpretation and readability. Robust standard errors in parentheses. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.



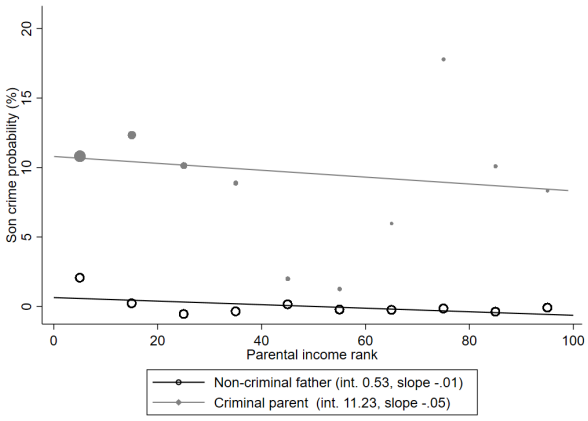
(a) Violent perpetration, sons



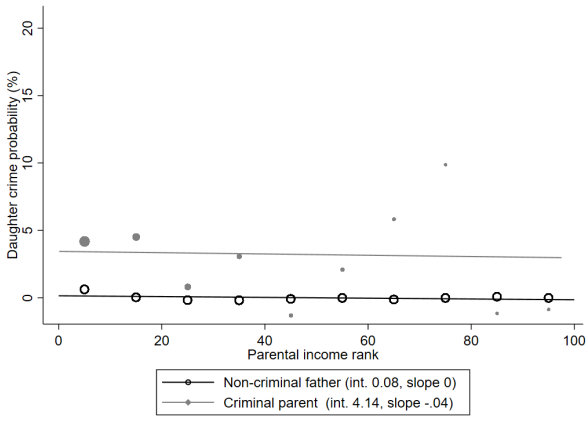
(b) Violent perpetration, daughters

Appendix Figure C2: Intergenerational transmission of violent perpetration

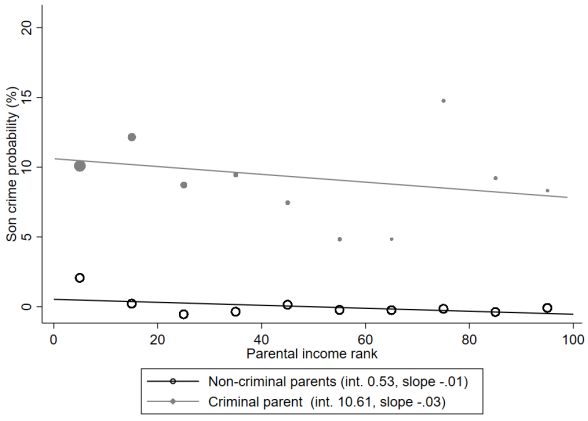
Note: These figures report estimates of violence transmission (γ_2) using equation (1). Violence outcomes are observed at child age 8-17 for parents, and child age 18-35 for children. The dependent variable is whether the child appears in the crime register for violence perpetration between the ages of 18 and 35. Income ranks (0-100) are determined within child birth-year cohorts, regardless of crime status. Point estimates (and standard errors) are rescaled by 100 to ease interpretation and readability. Robust standard errors in parentheses. 95% confidence intervals are reported.



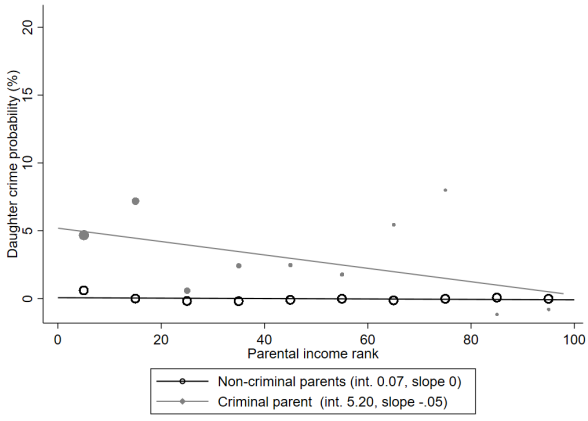
(a) Fathers to Sons



(b) Fathers to Daughters



(c) Any parent to Sons



(d) Any parent to Daughters

Appendix Figure C3: Child perpetration outcomes across parental income deciles by parental perpetration

Note: Each data point represents the perpetration rates at a given decile of the parental income distribution residualized with respect to all control variables except parental income. Violence outcomes are observed at child age 8-17 for parents, and child age 18-35 for the children. The dependent variable is whether the child appears in the crime register (for committed offenses) for a given crime between the ages of 18 to 35. Income ranks, 0-100, are determined within child birth-year cohorts, regardless of crime status. Cells with low counts are not reported for confidentiality reasons. Symbol sizes are weighted by the number of observations.

Appendix Table C4: Social mobility by parental violent crime perpetration and child gender

Dep. var: child's income rank				
	Daughters		Sons	
	(1)	(2)	(3)	(4)
Violent parent$_i$ (α_2)	-3.901***	-0.091	-9.158***	-2.290
	(1.339)	(1.332)	(1.607)	(1.593)
Parents' rank$_i$×	0.035	0.002	0.038	-0.042
Violent parent$_i$ (β_2)	(0.048)	(0.048)	(0.059)	(0.057)
Parents' rank$_i$ (β_1)	0.272***	0.151***	0.263***	0.155***
	(0.002)	(0.003)	(0.002)	(0.004)
Constant (α_1)	29.75***	25.32***	43.65***	33.28***
	(0.124)	(5.672)	(0.147)	(10.12)
Parental controls	No	Yes	No	Yes
Observations	182,031	182,031	188,010	188,010
R-squared	0.085	0.110	0.061	0.087

Notes: This Table reports results of rank-rank regressions allowing for heterogeneous effects of parental income by parental violent perpetration following equation (2). Violence outcomes are observed at child age 8-17 for parents. The dependent variable is the child's income rank where its income is averaged over ages 30-35. Income ranks, 0-100, are determined within child birth-year cohorts, regardless of perpetration status. Point estimates (and standard errors) are rescaled by 100 to ease interpretation and readability. Robust standard errors in parentheses. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table C5: Intergenerational transmission of violence for sons across types

Dep. var. (child):	Violent perpetration (1)	Violent victimization (2)	Domestic violence victimization (3)
Panel A: Any parent			
Victimized parent$_i$	3.89***	3.77***	2.08***
	(0.55)	(0.59)	(0.38)
Violent parent$_i$	9.08***	5.16***	2.08***
	(0.67)	(0.68)	(0.44)
Mean non-exposed	8.21	13.47	3.46
Observations	188,010	188,010	188,010
R-squared	0.057	0.029	0.019
Panel B: Fathers			
Victimized father$_i$	2.73***	2.79***	1.20**
	(0.68)	(0.72)	(0.46)
Violent father$_i$	8.94***	5.29***	2.17***
	(0.70)	(0.71)	(0.46)
Mean non-exposed	8.40	13.51	3.47
Observations	188,010	188,010	188,010
R-squared	0.055	0.028	0.018
Panel C: Mothers			
Victimized mother$_i$	6.16***	5.70***	3.63***
	(0.88)	(0.91)	(0.62)
Violent mother$_i$	11.88***	5.11*	2.56
	(2.12)	(2.08)	(1.43)
Mean non-exposed	8.45	13.77	3.58
Observations	188,010	188,010	188,010
R-squared	0.053	0.028	0.018
All controls	Yes	Yes	Yes

Notes: This table reports regression results of equation (1) with all control variables across violence types for sons. Violence outcomes are observed at child age 8-17 for parents, and child age 18-35 for children. The dependent variable is whether the child appears in the crime register for violence perpetration or in the emergency-room hospitalization data for assault victimization between the ages of 18 and 35. Point estimates (and standard errors) are rescaled by 100 to ease interpretation and readability. Robust standard errors in parentheses. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table C6: Intergenerational transmission of violence for daughters across types

Dep. var. (child):	Violent perpetration (1)	Violent victimization (2)	Domestic violence victimization (3)
Panel A: Any parent			
Victimized parent$_i$	1.28***	3.59***	2.32***
	(0.27)	(0.45)	(0.35)
Violent parent$_i$	2.17***	2.84***	2.54***
	(0.34)	(0.51)	(0.41)
Mean non-exposed	1.07	4.32	2.01
Observations	182,031	182,031	182,031
R-squared	0.019	0.031	0.026
Panel B: Fathers			
Victimized father$_i$	1.06**	2.27***	1.66***
	(0.33)	(0.53)	(0.42)
Violent father$_i$	2.09***	2.83***	2.36***
	(0.35)	(0.52)	(0.42)
Mean non-exposed	1.12	4.35	2.03
Observations	182,031	182,031	182,031
R-squared	0.018	0.030	0.025
Panel C: Mothers			
Victimized mother$_i$	1.82***	5.80***	3.54***
	(0.47)	(0.76)	(0.60)
Violent mother$_i$	4.68***	4.27*	3.54*
	(1.34)	(1.73)	(1.46)
Mean non-exposed	1.13	4.52	2.15
Observations	182,031	182,031	182,031
R-squared	0.018	0.030	0.025
All controls	Yes	Yes	Yes

Notes: This table reports regression results of equation (1) with all control variables across violence types for daughters. Violence outcomes are observed at child age 8-17 for parents, and child age 18-35 for children. The dependent variable is whether the child appears in the crime register for violence perpetration or in the emergency-room hospitalization data for assault victimization between the ages of 18 and 35. Point estimates (and standard errors) are rescaled by 100 to ease interpretation and readability. Robust standard errors in parentheses. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table C7: Intergenerational transmission of violence estimates in previous literature

Paper	Type of violence	Data and sample	Set of controls	Point estimate (SE) [CI]
Hernandez-Leal et al. (2025)	IPV victimization	DHS, 16 African countries	Unconditional	0.259 [x1.89] (0.005)
			Survey FEs, Region×Rural FEs, Ethnicity FEs, Ethnicity×Region×Rural FEs, daughters' age and education, son-in-law's age and education, HH wealth index	Controls explain only 1/3 of persistence (estimate ≈ 0.17 in Fig. 1)
Collins (2025)	IPV victimization: (1) Any IPV (2) Emotional IPV (3) Less severe physical IPV (4) Severe physical IPV (5) Sexual IPV	DHS: 742,106 women from 105 surveys in 55 countries	Unconditional	Any IPV: 0.270 [x1.94] (0.002)
			Cluster FEs + controls (age dummies, schooling, job status, HH wealth)	Any IPV: 0.168 [x1.59] (0.002)
Islam et al. (2014)	IPV victimization	DHS, Bangladesh	Unconditional Covariates	OR = 2.6 [2.2-3.1] OR = 2.4 [2.0–2.8]
Puno et al. (2023)	IPV victimization and perpetration	DHS, women in the Philippines	Age-adjusted model	Victimization: OR = 2.4 [2.2–2.7] Perpetration: OR = 2.4 [2.1–2.8]
			Individual, household and husband characteristics	Victimization: OR = 2.1 [1.9–2.4] Perpetration: OR = 2.1 [1.9–2.4]
Hjalmarsson and Lindquist (2012)	Violent perpetration	Stockholm Birth Cohort Study (15,117 individuals born 1953) + Swedish Work and Mortality Database	Unconditional	Sons: OR = 2.481 (0.257) Daughters: OR = 3.017 (0.746)
Hjalmarsson and Lindquist (2013)	Violent perpetration	Swedish data; 1,258,826 male non-adoptees	Unconditional	Father: 0.113 (0.003) Mother: 0.124 (0.010)
Van de Weijer et al. (2014)	Violent offending	Dutch Transfive Study; 3,440 individuals over 3 generations	Unconditional	Father to Son: OR = 2.309 [1.309-4.071]